Late to College: How Late Bloomers Have Shaped the US Wage Structure*

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Abstract

We challenge the consensus that individuals hardly return to education by documenting that 10-15% of US college graduates obtained their degree at or after age 30. Late bloomers, due to their large and varying share across cohorts and their lower college wage premia, significantly impact the US wage structure. We first show that late bloomers reduce the impact of age-specific relative college supply on the college premium. Second, we show that late bloomers have slowed down the decline in the gender pay gap among college workers. Finally, late bloomers account for a quarter of the college premium flattening after 2000.

Keywords: educational choice, late bloomers, college share, wage returns
JEL codes: I21, I26, J31

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1 Introduction

There is a widely held assumption – in economics and elsewhere – that the vast majority of college graduates go to post-secondary education right after high school and finish their education in their early 20s. While most college graduates in the United States indeed follow this standard route (referred to as early college graduates), we show that around a fifth of college graduates complete their college education at or after age 30 (referred to as late bloomers). We further document that late bloomers have a lower college wage premium than standard graduates. Due to these facts late bloomers have significantly impacted the evolution of the US wage structure. We show that late bloomers (1) reduce the impact of age-specific relative college supply on the college premium, (2) have increased the measured gender pay gap among college workers by 5.8%, and (3) account for a quarter of the college premium flattening after 2000.

We make two empirical observations: first that a large share of college graduates are late graduates, and second that late graduates have lower college wage premia. We document that in the US the continued education of individuals into their 50s is a robust, persistent, and widespread phenomenon. For example, at age 29, only 16.6% of the 1941 birth cohort had a college degree; by age 65, this fraction had increased to 23.9%. The equivalent numbers for the 1971 birth cohort are 29.6% at age 29 and 36.1% at age 48. Overall, among individuals born between 1935 and 1970, around 23% of those who graduated from college by age 50 did so after age 30. We also show that the share of late bloomers varies non-monotonically across birth cohorts. In the cross-section, late bloomers account for between 10 and 15% of college graduates between 1985 and 2020. We calculate the aggregate share of late graduates, and document that they account for about a fifth of the aggregate increase in the college share. A further decomposition by gender and race reveals large differences between demographic groups. Second, we show that the college premium depends on the age at graduation. Late bloomers' college wage premium is about two thirds of early graduates', and the gap narrows with experience.

These observations imply that any analysis of the US wage structure that does not control for the changing share of late bloomers by age misses an important element of compositional change. To measure this change, we compute the share of late bloomers by age over time. We then quantify the impact of late bloomers on various aspects of the US wage structure. First, we show how the presence of late bloomers nuances the impact of the age-specific relative supply of college workers on the college premium. We revisit the seminal model by Card and Lemieux (2001) and include late bloomers

in their framework. We show that failing to account for them biases the elasticity of substitution across age groups; based on the 1979-2019 period this leads to an underestimation of the elasticity, which overstates the importance of age-specific relative college supply in the evolution of the college premium. Second, we demonstrate the impact of late bloomers on the gender pay gap among college workers and the recent flattening of the college premium. We show that accounting for the varying share of late bloomers the decline in the composition-adjusted college gender pay gap is 5.8% larger between 1985-2020. Third, we propose an adjustment to standard college premium measures that corrects for the changing share of late bloomers among college graduates. We show that late bloomers account for 27% of the flattening of men's college premium that began in the early 2000s, and 23% of women's.

Our findings relate to, and have important implications for several strands of literature. First, because late bloomers represent 15% of college graduates in 2020, they should be explicitly included in analyses of higher education. However, most papers in economics simply assume that all college graduates follow the standard route of finishing college right after high school without any delays (i.e., Mincer (1974), Becker (1994), Keane and Wolpin (1997), Lee and Wolpin (2006)). While some papers do mention the existence of late college graduates, they do not comment on it. Altonji et al. (2016) remark that about a quarter of students in their data graduate before 20 or after 24 and exclude them from their sample. Another example is Murphy and Topel (2016), where Figure 5 shows similar patterns as we do for the within-cohort increases in the fraction of college graduates for both women and men. An exception is Bauman (2016), who documents the within-cohort increase in educational attainment, but does not study its aggregate implications. In the demography and sociology literatures, the increase in the aggregate college share is typically attributed to successive cohorts obtaining more education, e.g., Ryder (1965), Mare (1995), Cheeseman Day and Bauman (2000). Similarly to us, several papers complement the cohort succession explanation by linking the rise in educational attainment to late graduates, also by gender and race, such as Jacobs and Stoner-Eby (1998), Wilson et al. (2011), and Denice (2017). The education literature addresses a related phenomenon: delayed college graduation via a lengthening 'time to degree' into individuals' late twenties and early thirties (Fitzpatrick and Turner (2008), Bound et al. (2012)). However, none of these papers quantify the contribution of late graduates to the aggregate increase in the college share, which we show to be about one fifth of the total change, nor do they study their impact on the wage structure.

Second, we show that the college wage premium depends on the age at graduation. Monks (1997) and Elman and O'Rand (2004) show that those who complete college at

a later age obtain smaller returns to education, while Ferrer and Menendez (2009) find the opposite in Canadian survey data. Griliches (1980) and Leigh and Gill (1997) do not find differences between the wage paths of students who interrupt their studies and those who do not, while Light (1995) finds that education wage boosts are smaller when there has been a discontinuity. Our paper complements this literature by documenting the lower returns for late bloomers and assessing how this combined with their large and varying share have impacted the wage structure.

The key contribution of our paper is that it highlights the importance of accounting for late bloomers when studying the US wage structure. Studies of the changes in the average college premium or college gender gap typically account for the changing composition of the workforce by age, gender, and race, but never account for the changing share of late bloomers. A consequence of this is that the measured changes in these wage gaps are confounded by changes in the share and age structure of late bloomers. Two specific examples of this are the persistence of the college gender gap and the flattening of the college wage premium in the US. Blau and Kahn (2017) document that the gender pay gap decreased at the bottom of the skill distribution since the 1980s, but remained stubbornly persistent at the top. Many controls have been added to wage regressions to explain this persistence (Meara et al. (2020)), and demand-side explanations have been put forward by Beaudry and Lewis (2014). We show that the varying share of late bloomers amplifies the college gender gap, offering a new supply-side explanation. Regarding the college wage premium, Autor et al. (2008) describe how it flattened since the early 2000s. Ashworth and Ransom (2019) also documents the flattening across cohorts. Beaudry et al. (2015) and Valletta (2018) show that demand-side explanations – job polarizarion and the declining demand for high skilled workers – contribute to the flattening. We highlight a different and novel mechanism driven by the changing share of late bloomers, which accounts for approximately a quarter of the flattening.

The remainder of the paper is organized as follows. In Section 2 we document the existence of late bloomers in the US and assess their contribution to the rise in the aggregate college share. In Section 3, we document that the college wage premium varies by age at graduation. Section 4 quantifies the role of late bloomers in the evolution of the US wage structure over time. Finally, Section 5 provides a summary and concludes.

2 Late bloomers

In what follows, we document that the continued education of individuals late in life is a robust, persistent, and widespread phenomenon.

2.1 College share within birth cohorts

We first use repeated cross-sectional data to calculate the share of college educated individuals within a birth cohort in each year. We use two types of cross-sectional data: the decadal American Censuses from 1960 to 2000 combined with the yearly American Community Survey (ACS) from 2001 to 2019, from IPUMS provided by Ruggles et al. (2020), as well as the Annual Social and Economic (ASEC) supplement of the Current Population Survey (CPS) for 1964-2020 also from IPUMS provided by Flood et al. (2022). By using these two data we leverage the very large sample sizes of the Census, and the annual information provided by the CPS.

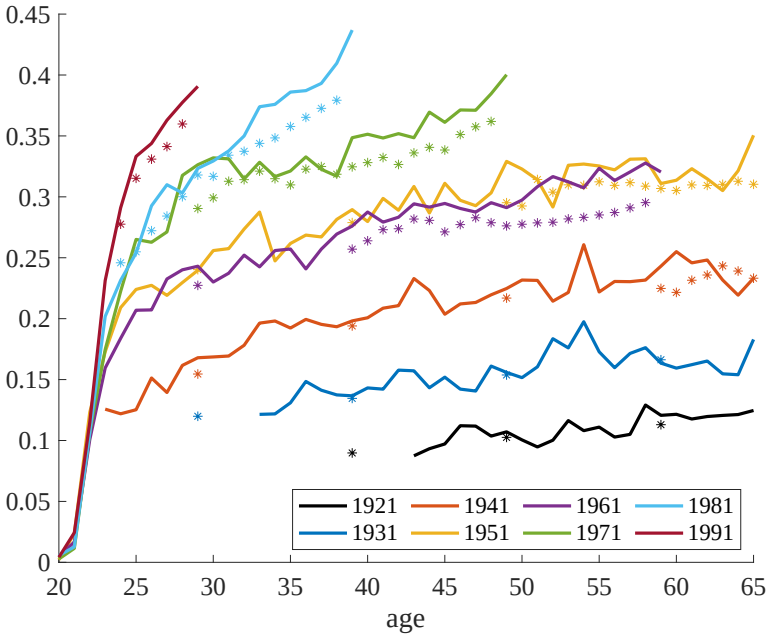


Figure 1: College share by age for different cohorts

Notes: Authors' calculations from Census/ACS (stars) and CPS ASEC data (solid curves). Each curve shows the share of college graduates for a birth cohort against age, calculated as the total sample weight of individuals from the birth cohort with at least 16 years of completed years of schooling divided by the total sample weight of individuals in the birth cohort.

Figure 1 plots the share of college-educated individuals for different birth cohorts between ages 20 and 65, where college education is defined as 16 years or more of completed years of education. The solid lines show the series from the CPS, and the stars show the college shares calculated from the Census/ACS. The striking feature of this

figure is that the college share increases substantially within each birth cohort as the cohort ages, even after age 29. According to the Census data, the college share at age 29 was around 15% in the 1941 birth cohort, and by age 65 it increased to around 24%; the equivalent numbers for the 1951 birth cohort are 24% and 31%, respectively. While a slight increase in the college share within birth cohorts is expected, the magnitude of this phenomenon is striking.[1] These patterns also hold for men and for women, and among the White and the Black population, see Figures B.1 and B.2 in the appendix.

The most natural explanation for the increasing share of college graduates within birth cohorts as the cohort ages is that individuals continue to acquire college degrees in their 30s, 40s, and 50s. While in repeated cross-sectional data we cannot ascertain whether individuals acquire education later in life, we verify that measures of school enrollment both from the Census and from the Integrated Postsecondary Education Data System (IPEDS) are in line with the increases in the college share within birth cohorts over age, see Section B.2 in the appendix for details.[2] The fact that the available data on enrollment is in line with the age gradient of the college share strongly suggests that a large share of college graduates obtain their degree later in life.

There are potential alternative explanations for the positive age gradient of the college share within birth cohorts. First, it is possible that the fraction of individuals with a college degree among immigrants is higher for older individuals, and that the age gradient of the college share of immigrants increased over time. We can rule this explanation out, as the same patterns hold when we consider only individuals born in the US, see Section B.3 in the appendix.

A second explanation could be that there are differences in the mortality rate of individuals with and without a college degree.A lower mortality rate among college educated workers would mechaincally increase the college share within a birth cohort. However, the available data suggest that the impact of mortality differences by education level on the evolution of the college share of a birth cohort is tiny. Therefore, we can rule out that mortality differences are driving the positive age gradient. See Section B.4 in the appendix for details.

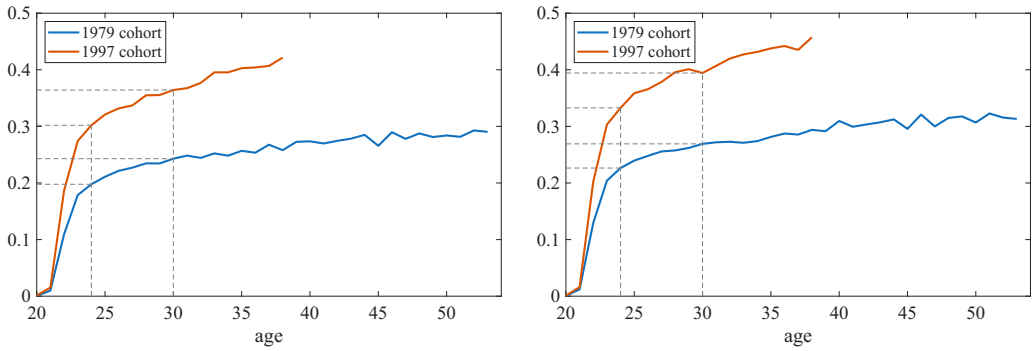
A third explanation is that perhaps the survey non-response rates are higher for those without a college degree, and this difference increases within a cohort as it ages. From repeated cross-sectional data we cannot rule out this explanation.

We next turn to panel data, where we can observe whether individuals acquire educa-

[1]The figure also confirms the well-known fact that educational attainment is higher for later-born cohorts (except for the 1961 birth cohort, which has a similar college share as the 1951 birth cohort at each age).

[2]Moreover, enrollment data from IPEDS and the Census are similar, which rules out that enrollment of older individuals in the Census is systematically mis-reported.

tion later in life and we can also rule out that non-response rates evolve differently by education as a cohort ages. We use the 1979 and 1997 release of the National Longitudinal Survey of Youth (NLSY) panel data set, which allow us to follow individuals over time from a young age. The NLSY data provides information on school enrollment, highest grade completed, employment, wages, as well as a rich set of descriptors. As in the cross-sectional data, we define college graduates as those who completed at least 16 years of schooling.



(a) Full sample (b) Stayer sample

Figure 2: College share by age in the two NLSY cohorts

Notes: Authors' calculations from the two releases of the NLSY. In each panel, the two curves show the share of college graduates against age for individuals from the NLSY79 (in blue) and the NLSY97 (in red). The left panel shows this in the full sample, whereas the right panel shows these shares among those who responded in all rounds of the respective survey.

In Figure 2, each curve represents individuals surveyed in one release of the NLSY. The blue curve, corresponding to the NLSY79, shows individuals born between 1957 and 1964, while the red curve, corresponding to the NLSY97, shows individuals born between 1980 and 1984. The left panel shows the college share at each age among all individuals surveyed at that age, while the right panel calculates the share at each age using a fixed sample, retaining only individuals who responded in all rounds of the survey. These graphs show that in both the NLSY79 and the NLSY97, individuals continue to acquire education as long as the survey follows them. Since the left and the right panel are similar, we can rule out that the age gradient in the college share is driven by differential survey non-response rates by education level. As in the cross-sectional, the same pattern holds for both men and women, as well as for all racial and ethnic groups, as shown in Figures B.7 and B.8 in the appendix. Therefore, the positive age gradient of the college share within cohorts, which we established using cross-sectional data, is also apparent in the NLSY. Moreover, the NLSY confirms that this is due to individuals acquiring education later in life.

In Figure 2 the age gradient in the college share becomes flatter around age 23-24, but after that age there is no clear breakpoint. This suggests that age 24 would be a natural candidate to separate late graduates from early. However, individuals graduating in their mid to late 20s might be just a little bit delayed, and markedly different from those who graduate later. We compare individuals in the NLSY79 data in terms of highest grade completed (HGC) and years of work experience accumulated in their 20s before college graduation. We find that these are similar among those who graduate before age 30, whether up to 24 or after, and they are also similar among those who graduate at or after age 30 or do not graduate at all during the sample period, see Appendix B.6 for details. Because of this and to be on the conservative side, in the rest of the paper we define late bloomers as those who obtained their college degree at or after age 30.

To get a sense of the differences between early, late and non graduates, in Table 1 we show socio-demographic descriptors of individuals at age 20 or earlier by age of college graduation in the NLSY79. Following our definition of late bloomers, we show this for early college graduates (who graduate before age 30), late college graduates (graduate at or after age 30), and those who do not graduate from college by the end of the sample period. The table shows that late graduates are distinct from both early

	early college	late college	non-college
Background characteristics			
HGC father	14.2	12.2	11.1
HGC mother	13.3	12.1	11.1
# of siblings	2.6	3.2	3.6
hh income	23,345	20,169	16,509
% living w parent at 17	81.1	66.1	59.1
% living w 2 parents at 17	64.7	38.8	40.2
Average skills (ASVAB)			
Coding Speed	52.4	49.1	40.5
Numerical Operations	40.0	37.8	30.3
Family status at age 20			
% married	3.0	23.2	24.3
% with children	6.3	51.7	61.9

Table 1: Background characteristics by age at college graduation

Notes: Authors' calculations from the NLSY79. Parent's highest grade completed (HGC) is computed in years of schooling. Household income is in $/year. The Armed Services Vocational Aptitude Battery (ASVAB) is a test administered to all individuals in the NLSY79 in 1980.

graduates and non-graduates in terms of their background characteristics, skills, and family status at age 20. Late graduates have lower parental education, come from

larger families, and have lower household income than early graduates. They are less likely to live with at least one of their parents at age 17. Late graduates are also more likely to be married and have children at age 20 than early graduates, but less likely than non-graduates. In terms of these demographic and family characteristics, late graduates are closer to non-graduates than to early graduates. Late graduates also have lower ASVAB scores than early graduates, but their average scores are closer to those of early graduates than of non-graduates. These observations suggest that late graduates have similar academic potential as early graduates, but they are more likely to come from disadvantaged backgrounds and have family responsibilities at a young age, which may delay their college graduation. They also provide further support for our definition of late bloomers at the cutoff age of 30 years old. We leave the investigation of the determinants of late graduation to future work, and focus on its aggregate implications in the remainder of this paper.

2.2 The share of late bloomers across cohorts and over time

We next calculate the share of late bloomers in each year, for which we need to observe the college share of each birth cohort at all relevant ages. Therefore, in what follows, we use the annual cross-sectional CPS data. We denote with $\lambda_{g,c,t}$ the college share of birth cohort c in year t for demographic group g. Birth cohort c's college share at age $\bar{a}$ is $\lambda_{g,c,c+\bar{a}}$. We calculate the share of late bloomers for birth cohort c in year t as the difference between the actual college share in year t and the share of those who graduated up to age 29:

$$\lambda^{\text{late}}_{g,c,t} \equiv \lambda_{g,c,t} - \lambda_{g,c,c+29}.$$

Note that for years $t \leq c + 29$ no individual from birth cohort c is more than 29 years old, and hence there cannot be any late bloomers, implying that $\lambda^{\text{late}}_{g,c,t} = 0$. The share of late bloomers among college graduates at age a in year t is given by

$$\mu_{g,a,t} \equiv \frac{\lambda^{\text{late}}_{g,t-a,t}}{\lambda_{g,t-a,t}}.$$

Table 2 shows these shares measured at age 50 averaged across different groups of birth cohorts in each row, and for different demographic groups of the cohort in each column. There are two main takeaways from this table. First, the share of late graduates at age 50 is between 14 and 47% across birth cohorts, but it is higher among female than male, and among Black than White college graduates. Second, it is non-monotonic across birth cohorts, decreasing after the early 1930s birth cohorts and then

birth cohorts	all	men	women	Whites	Blacks
1935-1939	30.0	28.7	30.5	27.6	47.2
1940-1944	24.5	19.4	30.6	23.2	46.3
1945-1949	21.0	16.7	26.1	21.6	27.9
1950-1954	21.3	16.3	26.7	20.2	28.3
1955-1959	18.0	13.9	22.1	16.8	29.2
1960-1964	20.4	19.7	21.1	17.8	37.2
1965-1970	23.4	19.4	26.8	22.2	37.6

Table 2: Share of late college graduates among college graduates

Notes: Authors' calculations from the CPS 1964-2020. The table shows the share of those who obtained a college degree at or after the age of 30 among those who obtained a college degree by age 50 for different population groups in each column, averaged within birth cohort groups in each row.

increasing after the 1950s birth cohorts, where the change in direction is at slightly different times across demographic groups.

We next build on the cohort-analysis to calculate the share of late bloomers in the cross-section over time and quantify their contribution to the aggregate increase in the college share. We compute an aggregate late college share for demographic group g in year t among the population aged 22 to 50 from the cohort late bloomer share, $\lambda^{late}_{g,c,t}$, according to the following formula

$$\lambda^{\text{late}}_{g,t} = \sum_{a=30}^{50} \omega_{g,a,t} \lambda^{\text{late}}_{g,t-a,t},$$

where $\omega_{g,a,t}$ is the year t share of age a individuals within demographic group g and age range 22-50, i.e., such that $\sum_{a=22}^{50} \omega_{g,a,t} = 1$ for all t.[3] We define $\lambda_{g,t}$, the share of college graduates in the cross-section, similarly.

Figure 3 shows the aggregate college share $\lambda_{g,t}$ among the 22-50 year old population in solid blue, and the aggregate late college share $\lambda^{\text{late}}_{g,t}$ in solid yellow between 1985 and 2020.[4] The dashed yellow line shows the share of late graduates among college workers in the aggregate, calculated as $\mu_{g,t} = \lambda^{\text{late}}_{g,t}/\lambda_{g,t}$. Both the share of college graduates and the share of late college graduates increased during this period, though the latter at a slightly faster pace, as reflected by the increase in the share of late graduates among college workers from around 10% to 15%. Note that the share of late graduates

[3] Note that we only sum from $a = 30$, as $\lambda^{\text{late}}_{g,t-a,t} = 0$ for everyone below age 30.

[4] When considering the population up to age 50, we have a meaningful aggregate late college share series from the year 1985 onwards. If we were to calculate the aggregate college share based on a wider age span, it would require us to consider a shorter time period. For reference, the dotted blue curve shows the aggregate college share among the 22-65 year-old population. Its level and its change is quantitatively very similar to that among the 22-50 year-old population.

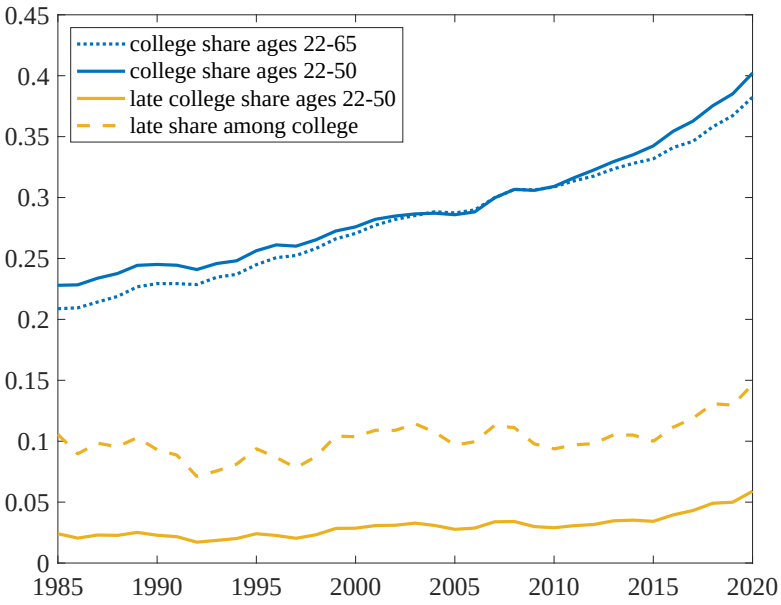


Figure 3: Late bloomers over time

Notes: Authors' calculations from the CPS 1964-2020. The blue curve shows the actual college share in the 22-50-year-old population between 1985 and 2020 (in dotted blue is the college share among the 22-65-year-old population). The yellow solid curve shows the share of late college graduates in the 22-50-year-old population.

among college workers in the aggregate is lower than its age-50 value for each cohort. This is because in the cross-section many cohorts are younger, implying a lower share of late graduates in those cohorts.

Figure 3 allows us to quantify the contribution of late bloomers to the aggregate increase in the college share, and relate it to the share of late bloomers among college graduates. The college share in the 22-50 year old population increased from 20.9% in 1985 to 38.4% in 2020, a 17.5 percentage point (pp) increase. The share of late college graduates increased from 2.4% to 5.9%, a 3.5 pp increase. This implies that 20% (3.5/17.5) of the increase in the aggregate college share in the 22-50 year old population is due to late graduates between 1985 and 2020, whereas on average over this period only 10.1% of college graduates were late bloomers. Table 3 below shows these numbers also for men, for women, for Whites and for Blacks.

This table shows that late bloomers contribute much more to the overall increase in the college share than what their average share among college graduates would imply. This is because in the population as a whole, and in all considered demographic groups, the share of late college graduates has increased faster over time than the share of college graduates. In the population as a whole, they contribute 20%, whereas their share on average was 10.1%. The contribution of late bloomers is higher than this for men at 25% (where their share is even lower), and for Blacks at 22.8% (where their

	aggregate	men	women	Whites	Blacks
Δ college share	17.5 pp	11 pp	23.7 pp	16.3 pp	16.9 pp
Δ late college share	3.5 pp	2.7 pp	4.3 pp	3.2 pp	3.9 pp
% late among college	10.1%	9.4%	11.1%	9.4%	18.5%
role of late bloomers	20%	25%	18.1%	19.4%	22.8%

Table 3: Contribution of late bloomers to aggregate college share increase

Notes: Authors' calculations from the CPS 1964-2020. The first row shows the percentage point (pp) increase in the share of college graduates ($\lambda_{g,2020} - \lambda_{g,1985}$), the second row shows the pp increase in the share of late college graduates ($\lambda^{\text{late}}_{g,2020} - \lambda^{\text{late}}_{g,1985}$), both among the 22-50 year old population between 1985 and 2020. The third row shows the period's average share of late bloomers among college graduates. The last row quantifies the share of the increase that is due to late graduates. Each column is for a different demographic group.

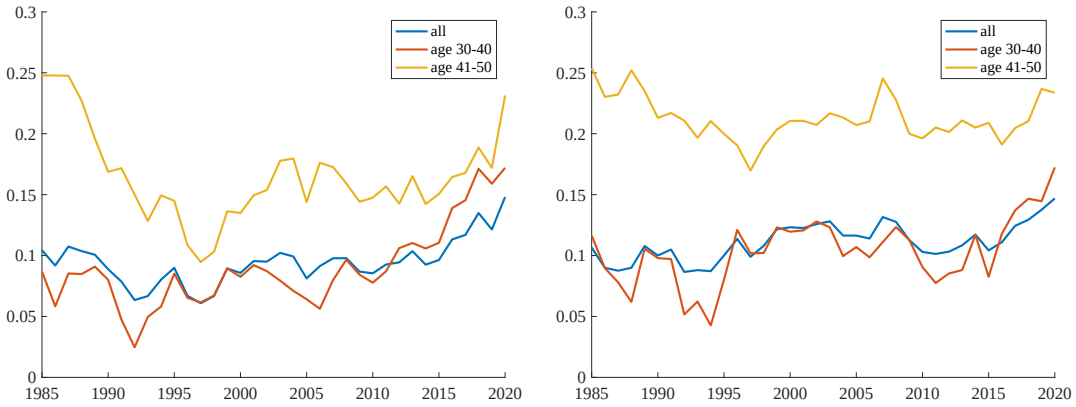
share is higher).

Finally, because the share of late bloomers varies non-monotonically across cohorts, they are likely to not be equally represented in each age group. This matters for the wage structure, as different age groups have different wage profiles. We therefore examine the share of late bloomers among college graduates by age group over time. The share among college graduates in demographic group g in year t between age a_1 and a_2 is computed as

$$\frac{\sum_{a\in[a_1,a_2]} \omega_{g,a,t}\lambda^{\text{late}}_{g,t-a,t}}{\sum_{a\in[a_1,a_2]} \omega_{g,a,t}\lambda_{g,t-a,t}}.$$

In Figure 4 we show this for men in the left panel and women in the right panel for all ages (ages 30-50) in blue, and for the younger (age 30-40) in red, and for older (age 41-50) college graduates in yellow. For men the overall share of late bloomers was stable around 10% until the early 2000s, followed by a 5 pp increase. For women the overall share of late bloomers among college graduates was steadily increasing throughout this period from around 10 to 15%. This relative stability overall masks differential changes across age groups. For both genders, the share of late bloomers among college graduates initially declines in the older age group. This decline is especially stark for men. Starting from mid-2000s, the share of late bloomers among college graduates starts to increase for both genders, especially in the younger age group.

In this section we documented that a substantial share of college graduates are late bloomers within each birth cohort and also in the cross-section of the population. We also documented that the share of late bloomers among college graduates has been increasing between 1985 and 2020, as well as that they significantly contribute to the increase in the aggregate college share. We also showed that the share of late bloomers among college graduates varies over time differently across age groups.



(a) Men (b) Women

Figure 4: Late bloomer share among college by age

Notes: Authors' calculations from the CPS 1964-2020. Both panels show the share of late bloomers among college graduates over time, by age group, and separately by gender. The blue curve shows the share over the 30-50 age group, the yellow curve shows the share over the 30-40 age group, and the red curve shows the share over the 41-50 age group.

3 The college wage premium by age at graduation

In this section, we estimate the college wage premium, i.e., the average wage of college workers relative to the average wage of non-college workers, by age at graduation. We conduct this analysis on the NLSY data, as we need to observe the age at which the individual obtained a college degree. Our main categories of interest are late versus early graduates, yet we acknowledge that individuals who graduate before age 30 form a heterogeneous group, hence we distinguish them by whether they graduated before or after 24. We therefore split respondents in the NLSY79 into four groups: (i) early college, those who graduate from college at or before age 24; (ii) medium college, those who graduate from college between age 25 and 29; (iii) late college, those who graduate from college at or after age 30, and (iv) never college, those who did not obtain a college degree during the sample period.

To estimate the college wage premium by age at graduation, we run the following wage-experience regression separately for each education group:

$$\log w_{it} = \alpha + \beta_{1,n}\text{exp}_{n,it} + \beta_{2,n}\text{exp}^2_{n,it} + \theta D_{it} + \beta_{1,c}\text{exp}_{c,it} + \beta_{2,c}\text{exp}^2_{c,it} + \gamma' X_i + \epsilon_{it}, \quad (1)$$

where $\log w_{it}$ is individual i's real log hourly wage in year t.[5] We use the panel aspect of the NLSY79 in constructing $\text{exp}_{n,it}$ and $\text{exp}_{c,it}$ which are individual i's labor mar-

[5]This is a generalization of the Mincerian regression as presented by Card (2001), where the additivity assumption between years of schooling and experience is relaxed and potential experience before and after college graduation is allowed to yield different returns.

ket experience without and with a college degree, D_{it} is a dummy variable for having graduated from college, and X_i is a set of individual controls. We either control for individual fixed effects or control for gender, race, skills (as measured by ASVAB scores) and other fixed characteristics.[6] By running the regression separately for each group, we can gauge whether the wage profile of the medium and of the late graduates is similar to that of the never graduates before obtaining a college degree, and whether it is similar to the early graduates once they obtain their degree. The college wage premium at a given experience level is then the difference in predicted log wages between the relevant college group and the never college group at that experience level.

Figure 5 depicts the trajectory of a representative individual from each of the four groups.[7] We compute this trajectory for each group by setting the covariates' values at their group's averages, and by assuming that the college degree is obtained at the average of the maximum years of non-college experience across all individuals in the group.[8] As expected, the never-college group has the lowest wage trajectory, while the early college group has the highest. The medium college group's wage trajectory after graduation is quite close to the early college group's, though stays slightly below it. Late graduates earn significantly more than the never college and medium college groups before graduation. Upon graduation, their college premium is about two-thirds of the early graduates' premium, but their trajectory after graduation seems steeper than for all other groups, thereby gradually narrowing the wage gap with the early and medium college groups.[9] The wage paths for these four groups appear qualitatively similar when we run regression (1) by gender and by race, see Figures C.12 and C.13 in the appendix. We also run wage regression (1) in the NLSY97 data to see whether the patterns are similar for later birth cohorts. The results are qualitatively very similar, see Figure C.14 in the appendix. The results of these regressions clearly establish that late bloomers, even after graduation, earn less than early graduates.
An important consequence of observing individuals in the medium and late college groups switching from non-college to college status is that the estimated college fixed effect captures the true average return to having the degree, rather than selection effects. The regression for each of these two groups can be thought of as an AKM-style

[6] See Table C.1 in the appendix for more details and descriptive statistics by group.

[7] See Table C.2 in the appendix for the estimated coefficients.

[8] Here we show the results from our preferred specification, where we control for individual fixed effects. The results are qualitatively and quantitatively similar when controlling for time-invariant observable characteristics of individuals, see Figure C.10 in the appendix. This suggests that the observed individual characteristics capture almost all of the information contained in the individual fixed effects.

[9] The steeper trajectory might come from the quadratic experience assumption. When we allow each year of experience to have a different effect on wages, late bloomers' wage path stays below that of early graduates, see Figure C.11 in the appendix.

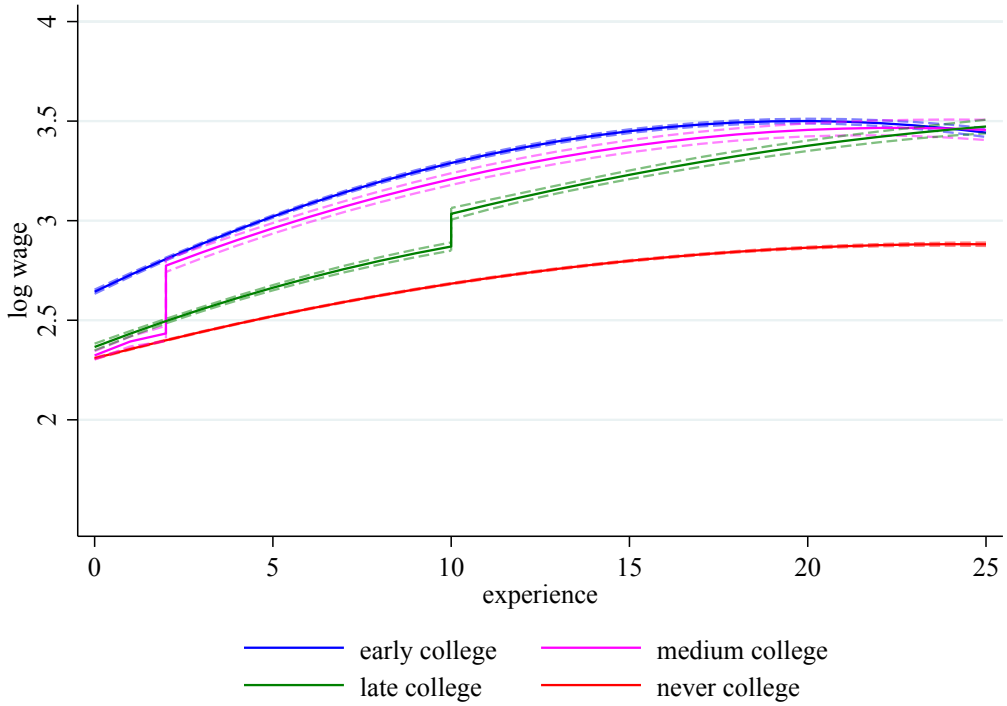


Figure 5: Average wage trajectories by group

Notes: Authors' calculations from NLSY79 data. Each curve shows the predicted wage path against experience from regression (1) for the representative individual from the early (blue), medium (pink), late (green) and never college group (red). The dashed lines show the 95% confidence intervals.

regression (Abowd et al. (1999)), with individual and firm/status fixed effects and only two statuses, non-college and college status. Since everyone in each of these groups eventually transitions from non-college to college status, but individuals move at different pre-college experience levels, we can identify the college fixed effect, i.e., the average individual wage gains from college graduation. This finding adds to the literature that seeks to estimate the causal effect of college education on earnings. See Griliches (1977) for a description of the issues arising from selection into higher education, and Card (1999, 2001) for surveys of the existing solutions, none of which exploit earnings observations before and after college graduation.[10] In this paper we do not aim to estimate causal college returns, but rather to show how differences in the college wage premium depending on age at graduation matter for the aggregate wage structure.

The difference in the estimated wage paths between early/medium and late college graduates is key for our further analysis. Whatever the reason for the difference in college wages of early and late graduates, it has implications for the US wage structure, and especially the college wage premium, defined as the difference between

[10]More recently, Zimmerman (2014) exploits a natural experiment to estimate the returns to college admission, and Mountjoy (2022) develops an instrumental variable approach to estimate heterogeneous returns to two-year community colleges.

the average college and non-college wage. In the next section, we show how late bloomers influence the wage structure through two main channels. The first is a direct channel driven by late bloomers' lower productivity—and correspondingly lower wages—relative to early graduates. The second is a composition channel arising from changes in the share of late bloomers among college graduates by age over time.

4 The impact of late bloomers on the wage structure

Thus far, we have established that (1) late bloomers constitute a substantial and time-varying fraction of college graduates; and (2) the wage profile of late graduates is lower than that of early graduates. These two facts jointly imply that late bloomers have an impact on the wage strcuture. We therefore revisit some of the findings and analyses of this literature. First, we examine how failing to account for late bloomers in the analytical framework of Card and Lemieux (2001) biases the estimates of the elasticity of substitution across age groups. Second, we quantify the contribution of late bloomers to the decline of the gender pay gap. Finally, we analyze the impact of late bloomers on the flattening of the skill premium in the US that started in the early 2000s, as documented in Autor, Katz, and Kearney (2008) and Valletta (2018).

4.1 Revisiting Card and Lemieux (2001)

In their seminal work, Card and Lemieux (2001) argue that the rising college premium among younger men in the US in the 1980s and 1990s is driven by a falling supply of younger college-educated workers. They do so using a supply and demand model in which the college premium of a given age group depends on the relative supply of college-educated workers within that age group. In this subsection, we modify their model to take into account late bloomers, and show that ignoring late bloomers biases the estimates of the elasticity of substitution across age groups.

The framework in Card and Lemieux (2001) assumes that the aggregate quantity of high-school (respectively college) equivalent workers is the constant elasticity of substitution aggregate of high-school (respectively college) equivalent workers of different ages a:

$$H_t = \left(\sum_a \alpha_a H_{at}^{\eta} \right)^{\frac{1}{\eta}}, \qquad C_t = \left(\sum_a \beta_a C_{at}^{\eta} \right)^{\frac{1}{\eta}},$$

where the parameters α_a and β_a measure the relative productivity of age-a workers, and η is the elasticity parameter across age groups. Output is produced according to

the following constant elasticity of substitution production function:

$$Y_t = (\theta_{ht} H_t^{\rho} + \theta_{ct} C_t^{\rho})^{\frac{1}{\rho}}, \tag{2}$$

where θ_{ht} and θ_{ct} is the productivity of high-school and college equivalent workers, and ρ is the elasticity parameter across skill groups. Assuming that labor markets are perfectly competitive, the college premium of age-a workers can be expressed as:

$$\frac{w_{at}^c}{w_{at}^h} = \frac{\theta_{ct}}{\theta_{ht}} \left(\frac{C_t}{H_t}\right)^{\rho-\eta} \frac{\beta_a}{\alpha_a} \left(\frac{C_{at}}{H_{at}}\right)^{\eta-1}. \tag{3}$$

We extend this framework based on the observation that a substantial share of college graduates are late bloomers, and that late bloomers' wages are lower than those of early graduates. We denote the quantity of age a college-equivalent labor at time t with $\tilde{C}_{at}$, and assume that it is itself an aggregate of age-a early E_{at} and late graduates L_{at} (note that $L_{at} = 0$ for $a < 30$ by construction):

$$\tilde{C}_{at} = [E_{at}^{\nu} + \gamma_a L_{at}^{\nu}]^{\frac{1}{\nu}}.$$

In the above, $\gamma_a < 1$ captures the relative productivity of late graduates compared to early graduates, assumed to be constant over time. This allows for the wages of early and late graduates of the same age to differ, mirroring our findings in the previous section. In what follows, for expositional purposes we assume that $\gamma_a = \gamma$, i.e., the relative productivity of late bloomers does not change over age, and that $\nu = 1$, i.e., early and late graduates are perfect substitutes.[11] Under these assumptions, their wages satisfy $w_{at}^{cl} = \gamma w_{at}^{ce}$, where w_{at}^{ce} and w_{at}^{cl} denote the wage in year t of age-a early and late college workers respectively. Average college wages among age-a individuals is then given by

$$w_{at}^c = \frac{E_{at} w_{at}^{ce} + L_{at} w_{at}^{cl}}{E_{at} + L_{at}} = w_{at}^{ce} \frac{E_{at} + \gamma L_{at}}{E_{at} + L_{at}}.$$

Note that the total amount of college equivalent workers is $C_{at} = E_{at} + L_{at}$, but this simple summation ignores the relatively lower productivity of late bloomers. Plugging in $\tilde{C}_{at}$ for C_{at} in (2), and combining the first order conditions for high school workers and early college graduates with the above expression for average college wages, we

[11] Note that the result of biased estimates for η when late bloomers are ignored is maintained even if γ_a varies with a and if $\nu \neq 1$.

obtain

$$\frac{w_{at}^c}{w_{at}^h} = \frac{\theta_{ct}}{\theta_{ht}} \left(\frac{C_t}{H_t}\right)^{\rho-\eta} \frac{\beta_a}{\alpha_a} \left(\frac{E_{at}+L_{at}}{H_{at}}\right)^{\eta-1} \left(\frac{E_{at}+\gamma L_{at}}{E_{at}+L_{at}}\right)^{\eta}. \tag{4}$$

Equation (4) is almost identical to the original (3), except for the last term $[(E_{at} + \gamma_a L_{at})/(E_{at}+L_{at})]^{\eta}$, which captures the presence and lower productivity of late bloomers. Card and Lemieux (2001) take the log of (3) and estimate the age substitution parameter η from

$$\log \frac{w_{at}^c}{w_{at}^h} = d_t + b_a - (1-\eta) \log \frac{C_{at}}{H_{at}} + e_{at}, \tag{5}$$

where d_t and b_a are time and age fixed effects, and e_{at} is the normal error term. Equation (4) suggests the following modification of (5) to account for late bloomers

$$\begin{aligned} \log \frac{w_{at}^c}{w_{at}^h} &= d_t + b_a - (1-\eta) \log \frac{C_{at}}{H_{at}} + \eta \log \frac{E_{at}+\gamma L_{at}}{E_{at}+L_{at}} + e_{at} \\ &\approx d_t + b_a - (1-\eta) \log \frac{C_{at}}{H_{at}} - \eta(1-\gamma)\mu_{a,t} + e_{at}, \end{aligned} \tag{6}$$

where $\mu_{a,t} = L_{at}/(E_{at} + L_{at})$ is defined as in Section 2. Comparing the two equations, the term $\mu_{a,t}$, the share of late graduates among age-a college workers is an omitted variable in Card and Lemieux (2001)'s estimating equation (5). As $\gamma < 1$, the relative wages w_{at}^c/w_{at}^h and the share of late bloomers $\mu_{a,t}$ are negatively correlated. The sign of the bias on the estimate for η therefore depends on the correlation between $\mu_{a,t}$ and $\log(C_{at}/H_{at})$. It could be positive if the share of late bloomers is higher in age groups with a higher relative supply of college-educated workers, and negative otherwise. In the first case, the bias would lead to an underestimation of η, and in the second case to an overestimation.

To get a sense of the magnitude of the bias in practice, we estimate both the original (5) and our modified version (6) on the ASEC CPS 1964-2020 data. We consider only cohorts born in or after 1935, which we can observe at age 29 allowing us to compute the share of late bloomers among college graduates. We follow Card and Lemieux (2001) as closely as possible in calculating high school and college equivalent labor supplies by age and year and in the calculation of the college premium by age and year. In addition, we compute $\mu_{a,t}$, the share of late bloomers among college workers by age and year, based on $\lambda_{g,c,t}$ and $\lambda_{g,c,t}^{\text{late}}$ as defined in Section 2.[12]

Table 4 shows the estimated coefficients when running the two regressions separately.

[12] See appendix Section D.1 for details.

	Card-Lemieux	Late bloomers
$\log \frac{C_{at}}{H_{at}}$	-0.233***	-0.195***
	(0.031)	(0.033)
$\mu_{a,t}$		-0.155***
		(0.057)
$\widehat{\eta}$	0.767	0.805
age-elasticity	4.30	5.14
$\widehat{\gamma}$		0.808

Table 4: Estimates of the age-elasticity with late bloomers

Notes: Authors' calculations from the ASEC CPS 1964-2020 data. The table shows the point estimate(s), the implied age-elasticity and the relative efficiency of late bloomers based on regression (5) in the first column, and based on regression (6) in the second column estimated for years 1979-2020.

We obtain estimates for η of 0.767 and 0.805 for the regressions with and without the late bloomers term. The implied estimate of γ is 0.81, suggesting that late bloomers are approximately 20% less productive than early graduates. The estimate of η and the implied elasticity of substitution between age groups (4.3 vs 5.14) is higher when accounting for late bloomers. A higher elasticity of substitution across age groups implies that the age-specific relative supply has a lower impact on the age-specific college premium. This shows that late bloomers are an important source of bias in the estimation of the age elasticity of substitution.

4.2 Implications for the gender pay gap

Next, we quantify the contribution of late bloomers to the evolution of the college gender pay gap. We do so by computing a counterfactual average college wage for both genders, where the only source of variation over time is the changing share of late bloomers within each age group of college-educated workers, and both the education-age composition and log wages of each group are fixed.

We rely on the ASEC CPS 1964-2020 data to calculate the average employment shares of college workers across gender and age, $\overline{\omega}_{g,a,coll}$, and we use the share of late bloomers among college graduates by age and gender in each year $\mu_{g,a,t}$, as defined in Section 2. We run a version of our previous wage regression (1) for early (up to age 29) and late (after age 29) college workers by gender using the NLSY79 data. To map wages to age, we rely on years of potential non-college and college experience, that enter the regression non-parametrically, as dummy variables.[13] We obtain from these regressions log wages for early, $\log w_{g,a,ec}$, and for late college workers, $\log w_{g,a,lc}$, for each age and

[13]See Section C.7 in the appendix for details.

gender. The counterfactual average log wage for college workers by gender in year t is calculated as

$$w_{g,coll,t} = \frac{\sum_{a=22}^{a=50} \overline{\omega}_{g,a,coll} \left[(1 - \mu_{g,a,t}) \log w_{g,a,ec} + \mu_{g,a,t} \log w_{g,a,lc} \right]}{\sum_a \overline{\omega}_{g,a,coll}}.$$

In Figure 6 we plot the variation in the average college wage driven solely by the changing share of late bloomers within different age-gender groups among college-educated workers, holding wages and the age structure fixed. This is calculated as $w_{g,coll,t} - w_{g,coll,1985}$, for both genders.[14]

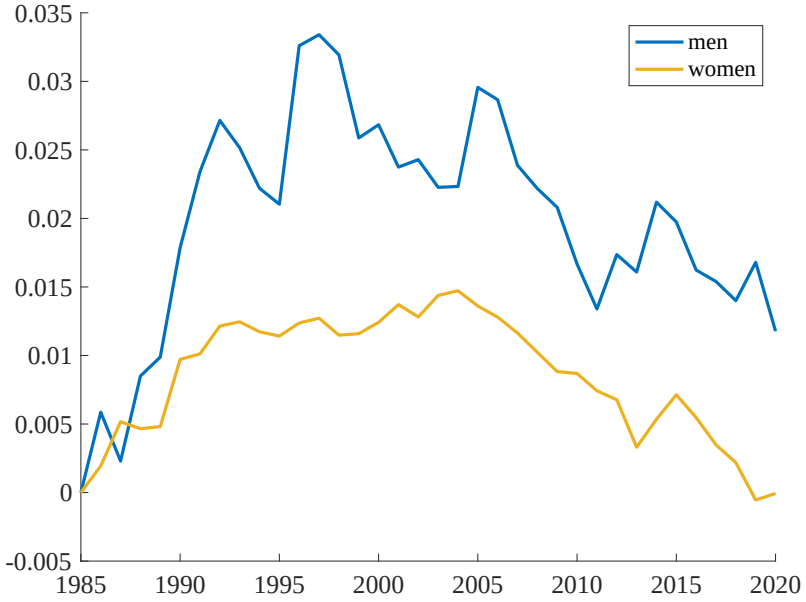


Figure 6: College wage variation due to late bloomers

Notes: Authors' calculations from the CPS ASEC 1964-2020 and NLSY79 data. Each curve shows the evolution of a counterfactual college wage among 22-50 year-olds, where the only source of variation over time is the changing share of late bloomers within different age-gender groups of college educated workers. The age x gender composition of the college educated workforce and the wage rates of each age x gender x early college/late college group are held fixed. The different colors show this for women and men separately.

The figure shows that the variation in late bloomer share pushed up the average college wage for both genders. This increase in hump shaped, reaching the highest point during the late 1990s/early 2000s. Throughout the 1985 to 2020 period the increase for

[14]Note that a similarly calculated average non-college wage by gender

$$w_{g,nc} = \frac{\sum_{a=22}^{a=50} \overline{\omega}_{g,a,nc} \log w_{g,a,nc}}{\sum_a \overline{\omega}_{g,a,nc}},$$

would be fixed over time by construction. This is because late bloomers do not impact the average non-college wage. This implies that to remove the variation in the college premium due to late bloomers we can also subtract the same measure, as $w_{g,coll,t} - w_{g,nc} - (w_{g,coll,1985} - w_{g,nc}) = w_{g,coll,t} - w_{g,coll,1985}$.

men was larger than for women, implying that the changing share of late bloomers exerted an upward pressure on the college wage gap between men and women. These observations are consistent with the variation by age group shown in Figure 4: the share of late bloomers among college graduates drops rapidly at the start of the period especially for men, which pulls up the college wage of men by more than of women. In the second half of the period, the late share among college increases for both genders in the 30-40 age group, which pulls down college wages.

Figure 7 allows us to quantify the contribution of late bloomers to the evolution of the college gender pay gap, which is defined as the difference between the log wage of college-educated men and women. In blue we show the composition-adjusted college gender pay gap, constructed as in Autor, Katz, and Kearney (2008), see Appendix D.2 for details. In orange we show its composition-and-late-bloomer adjusted version. We obtain this by removing the variation in the gender wage gap that is due to the changing share of late bloomers, i.e., we subtract the difference between the curve for men and the curve for women shown in Figure 6 from the composition-adjusted college gender wage gap.

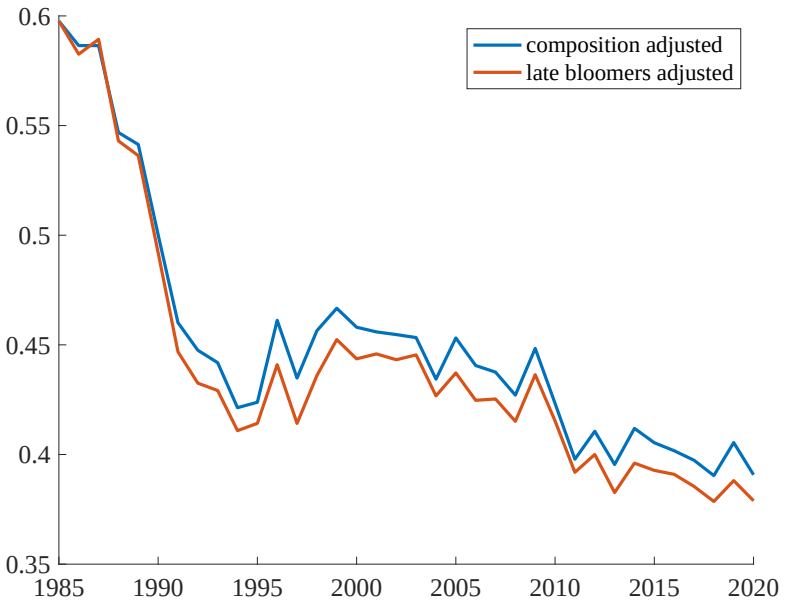


Figure 7: College gender pay gap unadjusted and adjusted for late bloomers

Notes: Authors' calculations from the CPS ASEC 1964-2020 and NLSY79 data. Each curve shows the evolution of a college gender pay gap among 22-50 year-olds. The blue line shows the standard composition-adjusted college gender pay gap, which lets the shares of late bloomers by age vary over time, while the orange line shows the college gender pay gap adjusted for the impact of late bloomers.

Between 1985 and 2020, the college gender pay gap dropped by 20.7 pp, while the adjusted college gender pay gap dropped by 21.9 pp. This shows that the changing

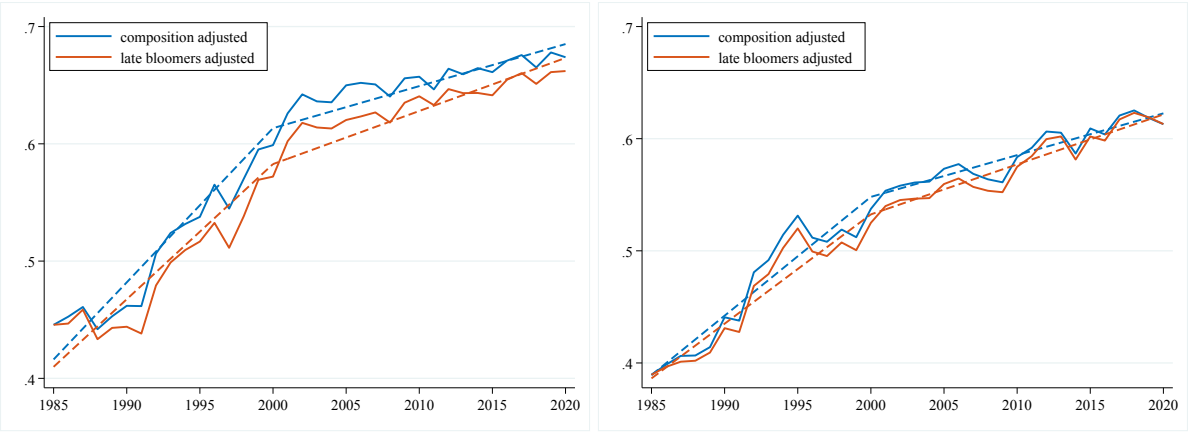
share of late bloomers among college-educated workers slowed down the measured decrease in the college gender pay gap by 1.2 pp over the period. This follows from Figure 6 as the variation in the late bloomer share increased men's college wage more than women's. As a result, once we fully adjust for composition effects, including for the share of late bloomers, the decline in the college gender pay gap is 5.8% larger.

4.3 Contribution to the flattening college premium

Next we turn to quantifying the impact of late bloomers on the college wage premium over time. For this note that the impact of the changing share of late bloomers among college-educated workers on the change in the college premium is exactly the same as their impact on the change in the average college wage. This is what is plotted in Figure 6, and it shows that the changing share of late bloomers exerted an upward pressure on the average college wage between 1985 and the late 1990s for both men and women, whereas from the early 2000s it pulled the average college wage downwards. The timing of this reversal in the direction of influence coincides with the flattening of the college wage premium in the US that started in the 2000s, as documented in Autor, Katz, and Kearney (2008) and Valletta (2018). This implies that the flattening of the fully composition-adjusted college premium, i.e., where also the share of late bloomers by age and gender is fixed, should be less pronounced.

Figure 8 shows the evolution of the composition-adjusted college premium (blue line) and of the composition-and-late-bloomers-adjusted college premium (orange line). The latter is the former adjusted by the impact of the changing share of late bloomers among college-educated workers by age across years. The figure for men is on the left, while that for women is on the right. The dashed lines show the fitted skill premia, assuming a structural break in 2001, to allow for the flattening.

The figure confirms that the increasing trend in the composition (but not late bloomer) adjusted college premium becomes flatter in the early 2000s, as pointed out by Autor et al. (2008). For men, the fitted slope drops from 1.3% to 0.3%, a 1 pp decrease, while for women it drops by 0.7 pp. The new observation is that, when adjusting for the changing share of late bloomers, the increase in the college premium until the early 2000s was less pronounced for both men and women, and the increase after the early 2000s was more pronounced. The decline in slope is therefore lower: 0.7 pp for men and 0.5 pp for women, which are, respectively, 73% and 77% of the unadjusted decline. This implies that the changing share of late bloomers is responsible for roughly a quarter of the observed flattening of the college premium for both genders. In other words, if the college premium were fully composition adjusted, i.e., it would account for the



(a) Men (b) Women

Figure 8: Effect of late bloomers on the college premium

Notes: Authors' calculations from the CPS ASEC 1964-2020 and NLSY79 data. The blue line depicts the college premium adjusted for composition in terms of 5 education groups, 2 genders, 3 races and 4 potential experience categories, but not adjusted for the share of late bloomers. The orange curve shows the late bloomer adjusted college premium. Dashed lines show fitted values. The structural break in 2001 is determined by a Supremum Wald test, using default symmetric trimming of 15% on the unadjusted college premium for both genders.

changing share of late bloomers, its flattening would be markedly less pronounced.

5 Conclusion

It is a generally accepted fact that most individuals who go to college complete their education in their early 20s. While previous research acknowledges that not all college graduates follow this pattern, this paper is the first to quantify the phenomenon's magnitude. We document that over the 1985-2020 period, between 10 and 15% of college graduates are late bloomers, who complete their college education in their 30s or even later. We map late bloomers by cohort, age and year in the US active population, and find that they drive a fifth of the increase in the aggregate college share between 1985 and 2020. This advocates strongly for a better awareness of late bloomers when designing educational public policy. Moreover, we show that late bloomers obtain a lower college wage premium than early graduates.

Most importantly, we demonstrate the quantitative importance of accounting for late bloomers in assessing the evolution and drivers of the college wage premium, and the college gender pay gap over time. We show that accounting for late bloomers reduces the estimated impact of the age-specific relative supply of college workers on the college premium. When accounting for late bloomers the decline in the gender pay gap among college workers is 5.8% larger. Finally, we demonstrate that late bloomers

account for approximately a quarter of the flattening of the college wage premium after 2000 for both men and women.

This paper highlights the need for explicitly accounting for late bloomers in research on college attainment and the college wage premium. While this task is difficult due to the lack of suitable data, it underscores the need for future datasets to record not only the highest degree completed, but also the exact year it was attained. Moreover, to complement the aggregate implications we study in this paper, a critical avenue for future research is identifying the structural and socioeconomic causes of late graduation.

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Appendix

A Data selection, variable definitions

We use the following Census and ACS data from IPUMS: 1960 1% sample, 1970 1% form 2 state sample, 1980 1% metro sample, 1990 1% metro sample, 2000 1% sample, and yearly ACS surveys (1%) for 2001-2019. We use the ASEC CPS data from IPUMS for the years 1964-2020. We keep individuals between 18 and 65 years old at the time of the survey and calculate their birth cohort from the year of the survey and age. In the Census and ACS we define education levels based on the detailed highest grade of school attended or completed variable *'higraded'* for the 1960-1980, and based on the detailed highest year of school or degree completed *'educd'* from 1990 onwards. College graduates are individuals who completed at least 4 years of college (*'higraded'*), or received a bachelor's degree or more (*'educd'*). In the ASEC data we rely on the highest year of school or degree completed *'educ'*, and define college graduates as those who completed at least 4 years of college, or received a bachelor's degree or more. We calculate the share of college-educated workers as the sum of person weights (*'perwt'* in the Census/ACS and *'asecwt'* in the ASEC CPS) among college-educated individuals divided by the sum of person weights among all individuals. We calculate this by birth cohort and year, and potentially one or more of the following categories: gender, race, foreign-born (only in the Census/ACS). We use the single race identification variable in the Census/ACS (*'racesing'*) and the race variable in the ASEC CPS (*'race'*) to determine whether an individual is White, Black, or other. Note that since the share of individuals in the 'Other' category is very small, and its encoding is not consistent over time, which increases its share time even within cohort, we only analyze trends for Black and White categories separately. To identify foreign and local-born individuals, we use the detailed birthplace variable from the Census/ACS (*'bpld'*) and code anyone born outside the US as foreign-born. We rely on the attending school variable from the Census/ACS (*'school'*) to determine whether an individual is attending school in a given year. We calculate the share of individuals attending school among all individuals using person weights by birth cohort, year, and also by level of completed education.

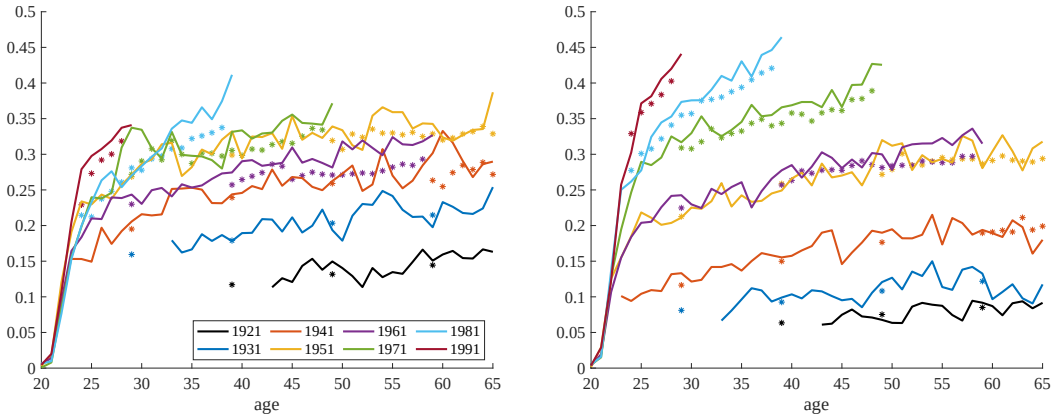
We use two releases from the NLSY: from 1979 to 2014 and from 1997 to 2019. Each release follows a cohort over time. We refer to these two cohorts as the NLSY79 and the NLSY97. Individuals in the same cohort may be of different ages: individuals are aged between 14 and 22 in 1979 in the NLSY79, and between 12 and 18 in the NLSY97. In the NLSY79 (NLSY97) interviews are conducted every year from 1979 to 1994 (1997-2013), and every two years from 1994 to 2014 (2013-2019). We exclude from the sample

individuals who are not interviewed every year for the first four years, as well as individuals who are interviewed less than four years in total. The NLSY79 and NLSY97 are not interviewed over the same time length: the NLSY79 covers 36 years, but the NLSY97 only covers 23 years. To compare the two cohorts we truncate the NLSY79 after 1998 to obtain comparable age distributions in the final year of the truncated NLSY79 and the NLSY97. We consider an individual a college graduate from the time they reach 16 years of education (if they do). We use the hourly wages provided in the surveys whenever available and construct them as yearly wages divided by the total number of hours worked in the year when necessary. To define years of labor market experience for the regressions in Section 3, we rely on the number of hours and weeks worked every year, as well as the number of months in which the individual reported to be at school or in college. All three variables are available also when interviews are bi-annual. An individual gains one year of experience in year t if the four following conditions are met in year $t-1$: they were at least 18, worked at least 800 hours, at least 25 weeks, and were in education (school or college) for less than 6 months. Non-college experience in a given year is labor market experience accumulated before the individual graduates from college, and college experience is accumulated after they graduate from college. College experience is set to zero in the year individuals graduate from college. Non-college graduates do not have any college experience by construction. We set early college graduates' non-college experience to zero. In section 3, we run the regression on individuals whose highest grade completed is known, who work, and who earn a nominal hourly wage above the federal minimum wage. We deflate nominal wages to perform the analysis on real wages. When performing the analysis on NLSY79 only, we cap observations at 25 years of total labor experience, for reasons of data scarcity. When performing the analysis on the truncated NLSY79 and NLSY97, we cap at 14 years of experience.

B Late bloomers

B.1 College share within birth cohorts for different demographic groups

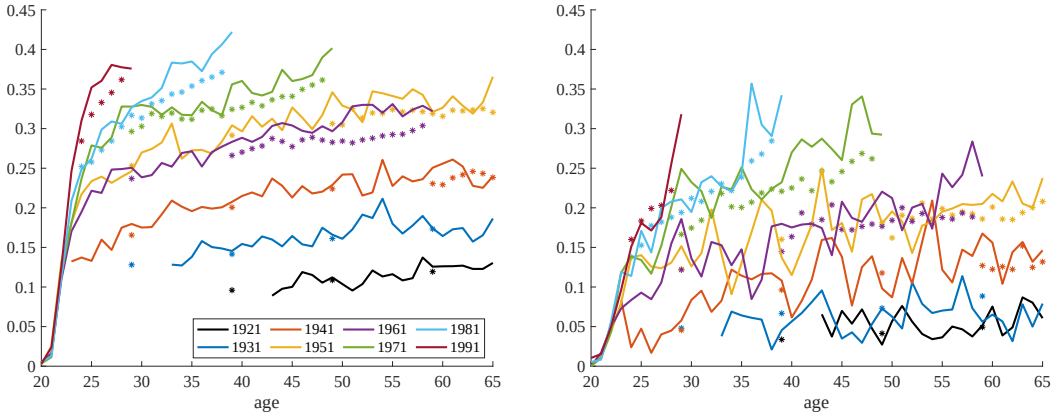
Figure B.1 shows that the same patterns hold for both men (on the left) and women (on the right) as in the aggregate. The two panels also show that the cross-cohort level differences in college share are much larger for women than for men. Among the cohorts born after 1971, the share of college graduates is much higher for women than for men. This shows how women caught up and overtook men in terms of college



(a) Men (b) Women

Figure B.1: College share by age and gender for different cohorts

Notes: Authors' calculations from Census/ACS (stars) and CPS ASEC data (solid curves). Each curve shows the share of college graduates for a birth cohort against age for men and for women.



(a) Whites (b) Blacks

Figure B.2: College share by age and race for different cohorts

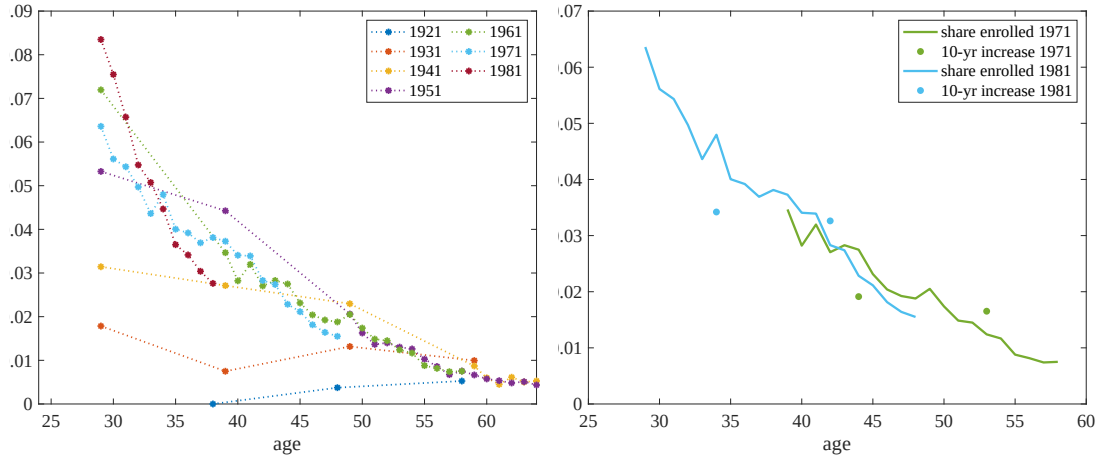
Notes: Authors' calculations from Census/ACS (stars) and CPS ASEC data (solid curves). Each curve shows the share of college graduates for a birth cohort against age by race.

education. Figure B.2 shows that these patterns hold by race as well.[15] The college share at age 29 is lower for Blacks than for Whites, but the increase over age is similar, suggesting that the share of late graduates is higher among Black college graduates.

[15]In our sample period the only race categories that can be consistently defined are White, Black, and other. The fraction of individuals in the other category is very small, and their share *within* a cohort is increasing over time, implying that the classification of the same individuals must have changed over time. For this reason, we do not include individuals in the other race category when breaking down the data by race.

B.2 School enrollment in the Census and in IPEDS

To verify whether individuals are getting education later in their life, we compare the available information on enrollment in school to the increases in the college share within birth cohort over age. The left panel of Figure B.3 shows the share of individuals enrolled in school among those high school graduates who do not have a college degree by birth cohort as they age. Since these individuals have at least graduated from high school but did not earn a college degree, they are attending college. It clearly demonstrates that a significant share of the 30, the 40, and even the 50 year-olds are enrolled in college and that these fractions increase across birth cohorts (the fraction at a given age in later-born cohorts is higher). The right panel of the same figure zooms in on the 1971 and the 1981 birth cohorts and on the years where yearly data is available. This figure shows the same college enrollment share curves as on the left, as well as 10-year increases in college attainment from age 29 to 39 and age 37 to 47 for the 1981 birth cohort, and from age 39 to 49 and age 48 to 58 for the 1966 birth cohort, at the midpoint of the 10-year intervals. The figure clearly shows that the measured 10-year increases in the college share are in line with reported school enrollment. However, because individuals surveyed by the Census self-declare both their



(a) College enrollment (b) 10-year change in college share

Figure B.3: College enrollment and 10-year increases in college share

Notes: Authors' calculations from the Census 1960-2010 and ACS 2001-2009 and 2011-2019. In the left panel each curve shows the share enrolled in school among those high school graduates who do not have a college degree, for a given birth cohort against age (same birth cohorts as in Figure 1). The right panel shows segments of these curves for the 1971 and the 1981 birth cohorts as well as 10-year changes in the college share of these birth cohorts at the midpoint of the 10-year intervals.

level of education and their school enrollment status, systematic misreporting skewed towards college attendance or graduation may occur. To check for this, we use data from the Integrated Postsecondary Education Data System (IPEDS), which is a system

of surveys conducted annually by the U.S. Department of Education's National Center for Education Statistics. We use data on school enrollment by age group, coupled with total population counts from the Census, between 2003 and 2021. The college enrollment shares by age group in the IPEDS is slightly above those from the Census/ACS. They nevertheless follow closely the reported school enrollment shares among those high school graduates who have not earned a college degree.

Figure B.4 shows the percentage of enrolled students of a given age bin among the total population in that age bin. The numbers for enrolled students are computed from IPEDS data, in which individuals do not self-declare but instead students are counted in colleges, as well as the ACS, which reports self-declarations. Enrollment as reported in the ACS is in line with the IPEDS, and both show that a significant share of individuals are enrolled in college after 25, even if it is slightly underestimated among the 25 to 39 age group in the ACS.

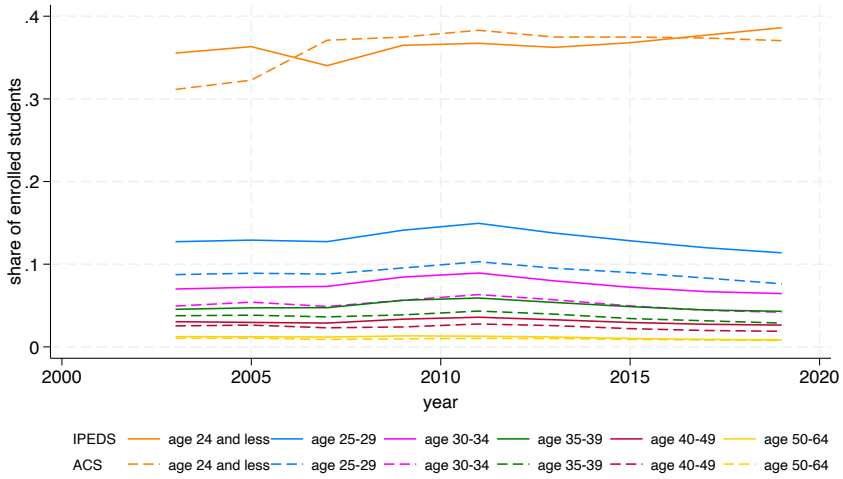


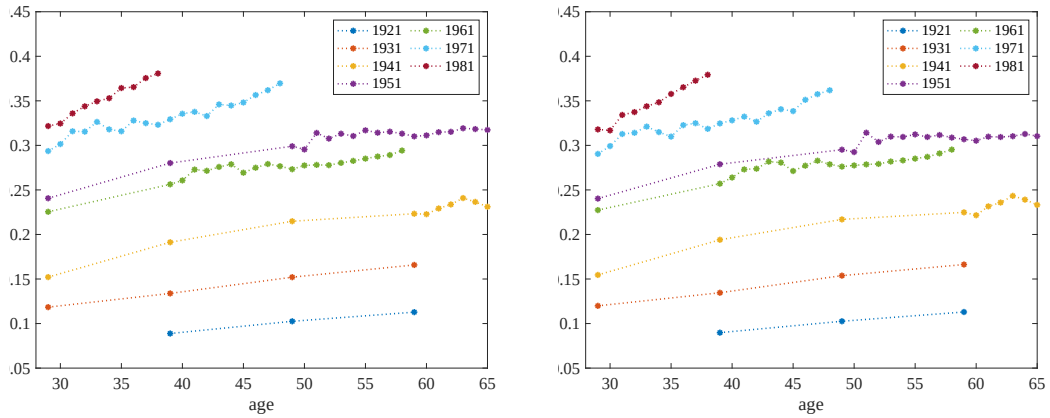
Figure B.4: Share of population enrolled in college by age bin

Notes: Authors' calculations from the IPEDS and ACS data from 2003 to 2021. Each curve shows the share of students enrolled in college, as a percentage of the total US population in that age bin, against years. In both data sources, the share is computed by taking the fraction of students enrolled according to that data source, divided by the total population in the age bin from the ACS data.

This suggests that the IPEDS data on enrollment is in line with the Census/ACS school attendance, which in turn is in line with measured increases in the college share as cohorts age. Thus it follows that these within-cohort increases in the college share are not due to a systematic misreporting of education levels as cohorts age.

B.3 College share by age and place of birth

Figure B.5 shows the share of college-educated individuals in the Census and ACS for different birth cohorts as they age, among local-born individuals (on the left) and in



(a) Local (b) Full sample

Figure B.5: College share by age and place of birth for different cohorts

Notes: Authors' calculations from the Census 1960-2010 and ACS 2001-2009 and 2011-2019. Each curve shows the share of college graduates for a birth cohort against age for local-borns and in the full sample.

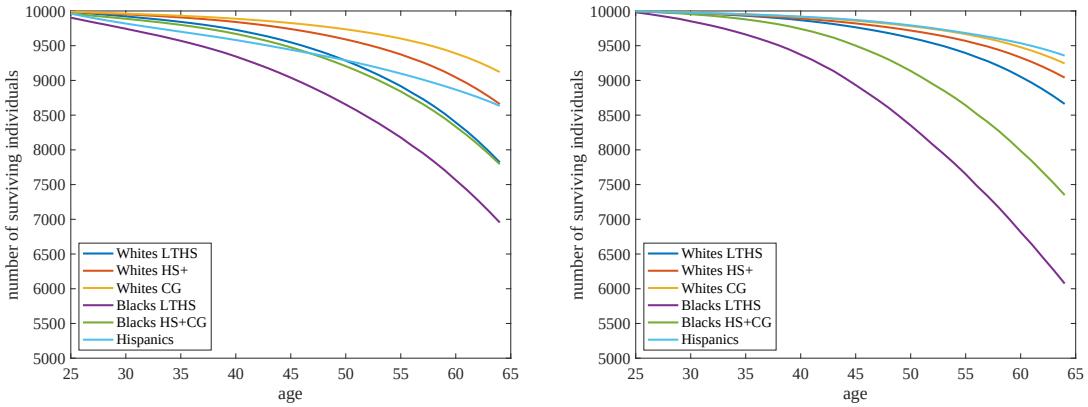
the full sample (on the right). The main takeaway is that the patterns we document hold among the local-born population, implying that the increasing share of college-educated individuals as a birth cohort ages is not due to immigration.

B.4 The impact of mortality differences

While there is some evidence that mortality is higher for individuals with lower educational attainment (see for example Roy et al. (2020)), the difference is not large enough to account for the patterns observed in the data. Figure B.6 illustrates the magnitude of mortality differences. This figure is computed using information from three different sources. First, we use the life tables from the National Vital Statistics Report for 2011, in which we have information about mortality rates by race. Second, we use the annual life table statistics provided by the National Vital Statistics Reports for all years. Finally, we use relative mortality rates by race, ethnicity, and education reported in Brown et al. (2002). These three sources make it possible to calculate the annual survival rate for several race and education groups by gender.[16] Figure B.6 shows the surviving population against age for the various education and race groups for men on the left and for women on the right, starting from a population of 10,000 individuals at age 25. We can see that there are large differences across the groups. Nevertheless, these differences are far too small to raise the percentage of college graduates to the extent that is observed in the data. To illustrate this, assume that at age 25 among White women 30% are college graduates, and among Black women 15% are college

[16]The resulting mortality rate tables by gender, education, and race are available on request.

graduates. The differential in survival rates after 20 years would lead to 30.6% college graduates among White women, and 15.6% among Black women. These are all very small changes relative to those observed in the data.

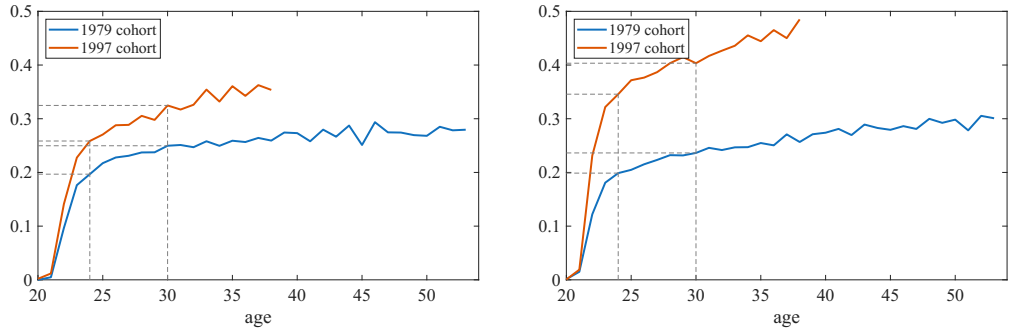


(a) Men (b) Women

Figure B.6: Simulated surviving population by gender, race and education

Notes: Authors' simulations starting from 10000 individuals at age 25 based on several data sources, as described in the text. The education categories are: less than high school education (LTHS), high school but without college (HS+), college degree (CG), high school or more (HS+CG).

B.5 College share by age for different demographic groups – NLSY

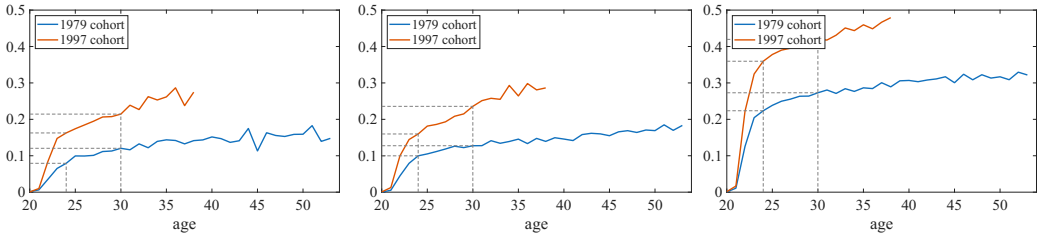


(a) Men (b) Women

Figure B.7: College share by age and gender in the two NLSY cohorts

Notes: Authors' calculations from the two releases of the NLSY. The curves show the share of college graduates against age for men and for women.

Figure B.7 shows that the same patterns hold both for men and for women, and that, as in the Census, the college share among women increased both due to a higher age 24 college share and due to a steeper age-gradient in educational attainment. Figure B.8 shows that these facts hold both among the Black, the Hispanic, and the non-Black non-Hispanic population.



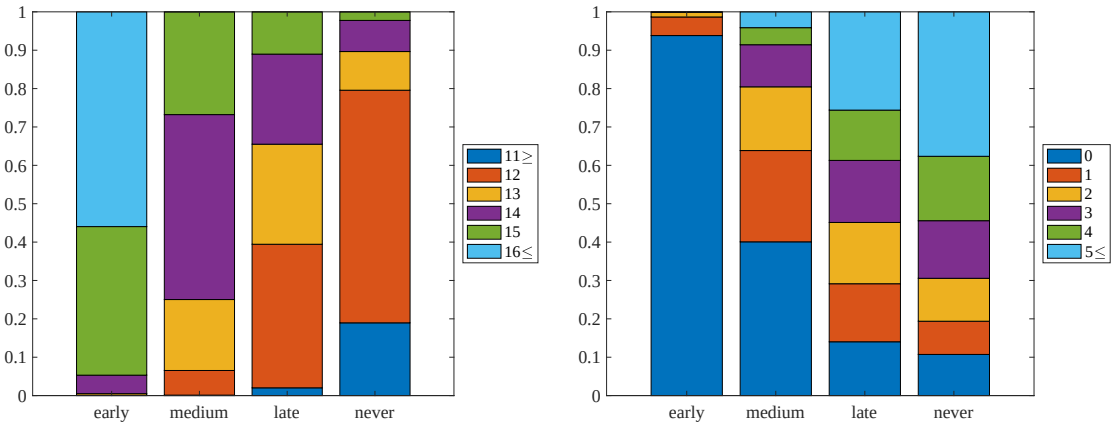
(a) Hispanics (b) Blacks (c) others

Figure B.8: College share by age and race in the two NLSY cohorts

Notes: Authors' calculations from the two releases of the NLSY. The curves show the share of college graduates against age for Hispanics, for Blacks, and for the non-Hispanic non-Black population.

B.6 HGC and work experience by college groups

Figure B.9 presents the distribution of HGC at age 22 for the 4 groups on the left, and on the right, it presents the distribution of pre-college work experience at age 26. At age 22, more than 95% of the early college group have at most one year left to complete college, and almost 75% of the medium college group have at most 2 years left. In stark contrast, almost 40% of the late college group and 80% of the never college group did not even start college. In terms of pre-college work experience, more than 90% of the early college group have none, and about 60% of the medium college group have at most 1 year. In contrast, around 60% of the late and 70% of the never college group have 3 years or more work experience by age 26.



(a) Highest grade completed at 22 (b) Work experience at 26

Figure B.9: Distribution of highest grade completed and work experience

Notes: Authors' calculations from NLSY79 data. The panel on the left shows the distribution of the highest grade completed (measured in years) at age 22 among those who end up in the early, medium, late, and never college groups. The right panel shows the distribution of pre-college work experience (in years) at age 26 for the same four groups.

C College wage premium by age at graduation

C.1 Descriptive statistics

Table C.1 presents the sample means of the variables used in the regression analysis for the NLSY79, the truncated NLSY79, and the NLSY97 cohorts. In the NLSY79, the difference in average maximum non-college experience is especially stark between medium and late college graduates: medium college graduates have little experience before they finish college (1.33 years), and accumulate college experience similarly to early college graduates (15.84 years and 17.59 years). Meanwhile, late college graduates accumulate much more non-college experience (10.01 years), and less post-college experience (8.89 years). This difference in human capital accumulation is the reason why we find it important to distinguish between these two groups. Similar observations can be made about the truncated NLSY79 and the NLSY97.

C.2 Regression results

Table C.2 presents the estimation results of equation (1) on the NLSY79, the truncated NLSY79, and the NLSY97. These are the estimated coefficients we use to construct the wage paths presented in Figures 5 and C.14.

	early	medium	late	never
			NLSY79	
# individuals	1678	354	476	8847
# observations	23795	6067	8657	124250
mean max exp non-college	-	1.33	10.01	15.80
mean max exp college	17.59	15.84	8.89	-
share women	0.53	0.49	0.65	0.48
share Black	0.16	0.22	0.33	0.27
share Hispanic	0.08	0.17	0.17	0.17
mean ASVAB coding speed	53.46	48.51	47.67	39.65
mean ASVAB numerical operations	40.77	36.86	36.48	29.69
mean weight (kg)	65.47	65.35	64.52	66.22
mean height (m)	1.72	1.71	1.69	1.70
mean BMI	22.11	22.19	22.50	22.80
median drug use	0	0	0	0
			NLSY79 truncated	
# individuals	1677	352	219	9049
# observations	17427	4573	2913	98075
mean max exp non-college	-	1.34	5.15	8.89
mean max exp college	9.24	6.59	2.83	-
share women	0.53	0.49	0.61	0.49
share Black	0.16	0.22	0.26	0.27
share Hispanic	0.08	0.16	0.16	0.17
mean ASVAB coding speed	53.48	48.59	48.82	39.93
mean ASVAB numerical operations	40.77	36.87	37.09	29.92
mean weight (kg)	65.46	65.30	65.31	66.19
mean height (m)	1.72	1.71	1.70	1.70
mean BMI	22.10	22.18	22.40	22.80
median drug use	0	0	0	0
			NLSY97	
# individuals	1886	443	372	4841
# observations	16938	5625	4336	55711
mean max exp non-college	-	1.78	4.91	8.53
mean max exp college	8.69	5.88	1.59	-
share women	0.59	0.54	0.58	0.45
share Black	0.16	0.24	0.33	0.30
share Hispanic	0.12	0.22	0.22	0.24
mean ASVAB coding speed	7801.34	6597.62	6607.70	5516.19
mean ASVAB numerical operations	19796.23	17382.86	17014.99	14645.12
mean weight (kg)	57.50	58.20	59.70	60.80
mean height (m)	1.65	1.65	1.64	1.65
mean BMI	21.04	21.43	22.04	22.34
median drug use	0	0	1	1

Table C.1: Descriptive statistics by college group

Notes: Authors' calculations from the NLSY79 and the NLSY97. The sample is the same as the one used in Table C.2, see Appendix A. The change in magnitude of the ASVAB scores between the NLSY79 and the NLSY97 is due to a different scaling of the test scores in the two cohorts.

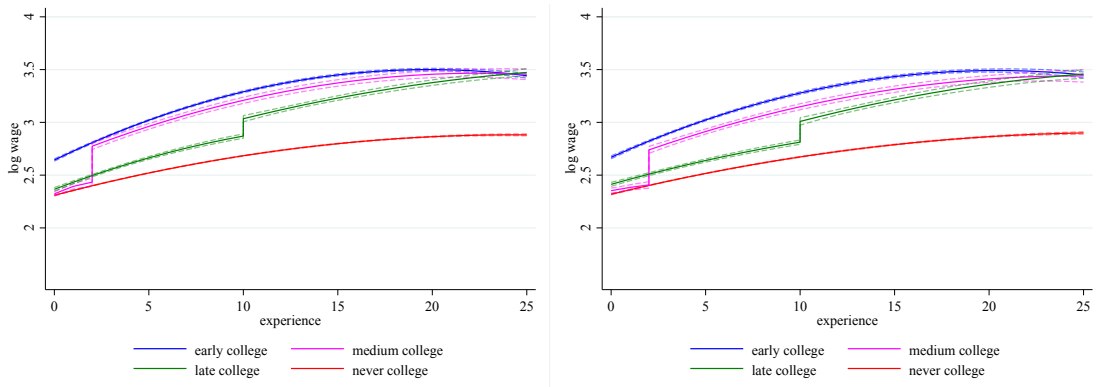
	log wage			
	early	medium	late	never
		NLSY79		
const	2.643	2.323	2.366	2.309
	(.005)	(.011)	(.009)	(.002)
$\exp_n$	-	.085	.069	.047
		(.016)	(.003)	(.000)
$\exp_n^2$	-	-.015	-.002	-.001
		(.003)	(.000)	(.000)
D	-	.341	.164	-
		(.017)	(.016)	
$\exp_c$	.086	.068	.044	-
	(.001)	(.003)	(.004)	
$\exp_c^2$	-.002	-.002	-.001	-
	(.000)	(.000)	(.000)	
# observations	23,795	6,067	8,657	124,250
# individuals	1,672	354	476	8,847
R^2	0.318	0.458	0.360	0.203
		Truncated NLSY79		
const	4.595	2.027	1.955	3.174
	(.495)	(.821)	(.989)	(.144)
$\exp_n$	-	.058	.089	.052
		(.013)	(.009)	(.001)
$\exp_n^2$	-	-.009	-.006	-.002
		(.003)	(.001)	(.000)
D	-	.335	.208	-
		(.019)	(.029)	
$\exp_c$	.104	.057	.012	-
	(.003)	(.009)	(.023)	
$\exp_c^2$	-.004	-.002	.004	-
	(.000)	(.001)	(.004)	
# observations	17,427	4,573	2,913	98,075
# individuals	1,671	352	219	9,048
R^2	0.235	0.346	0.223	0.126
		NLSY97		
const	4.087	.081	2.824	2.302
	(.377)	(.612)	(.636)	(.146)
$\exp_n$	-	.080	.093	.087
		(.012)	(.007)	(.002)
$\exp_n^2$	-	-.012	-.005	-.003
		(.002)	(.001)	(.000)
D	-	.371	.344	-
		(.020)	(.026)	
$\exp_c$	.104	.087	.027	-
	(.004)	(.010)	(.027)	
$\exp_c^2$	-.004	-.003	.000	-
	(.000)	(.001)	(.005)	
# observations	16,938	5,625	4,336	55,711
# individuals	1,869	443	372	4,841
R^2	0.187	0.352	0.259	0.220

Table C.2: Estimates from equation (1)

Notes: All regressions contain gender and race fixed effects, and individual fixed effects.

C.3 Individual time-invariant controls

Below we investigate whether the wage paths are similar when controling for individual time invariant characteristics, rather than using individual fixed effects. We control for gender, race, height, weight, and body mass index measured in 1981, and the number of times the individual used drugs in 1979, as well as for the individual's scores in Numerical Operations and in Coding Speed from the Armed Services Vocational Aptitude Battery (ASVAB) test completed in 1980.[17] Figure C.10 shows that controling for observables in the NLSY79 yields very similar results as including individual fixed effects. The college graduation premium is smaller with individual controls than fixed effects for the medium graduates, and the pre-college experience path is flatter. However, qualitatively the two graphs are identical and quantitatively they are very close to each other.[18]



(a) Individual fixed effects (b) Individual time-invariant controls

Figure C.10: Individual fixed effects vs time-invariant observable characteristics

Notes: Authors' calculations from the NLSY79. Each curve shows the predicted wage path (and 95% confidence interval) against experience from regression (1) for the representative individuals in the four college groups with individual fixed effects and with individual controls.

C.4 Flexible returns to experience

Figure C.11 shows the predicted wage paths for each college group from the following regression:

$$\log w_{it} = \alpha + \sum_{s=0}^{30} \beta_{s,n} \mathbf{I}\{\text{exp}_{n,it} == s\} + \sum_{s=0}^{30} \beta_{s,c} \mathbf{I}\{\text{exp}_{c,it} == s\} + \gamma' X_i + \epsilon_{it}. \quad (7)$$

[17] We choose to use specific ASVAB scores over AFQT scores as the latter are an aggregation of the former

[18] The wage paths by gender, by race/ethnicity, in the 1979 truncated and in the 1997 samples are similar when using time-invariant controls instead of individual fixed effects. These graphs are available on request.

This equation is very similar to (1) except that instead of assuming that experience affects wages in a quadratic way, we allow for each year of experience to have a different effect on wages. Note also that in this case the college completion dummy can be omitted, as it will be subsumed in the relevant $\beta_{s,c}$. The predicted wage paths are very similar to those in Figure 5, suggesting that the assumption of a quadratic effect of experience on log wages is not overly restrictive.

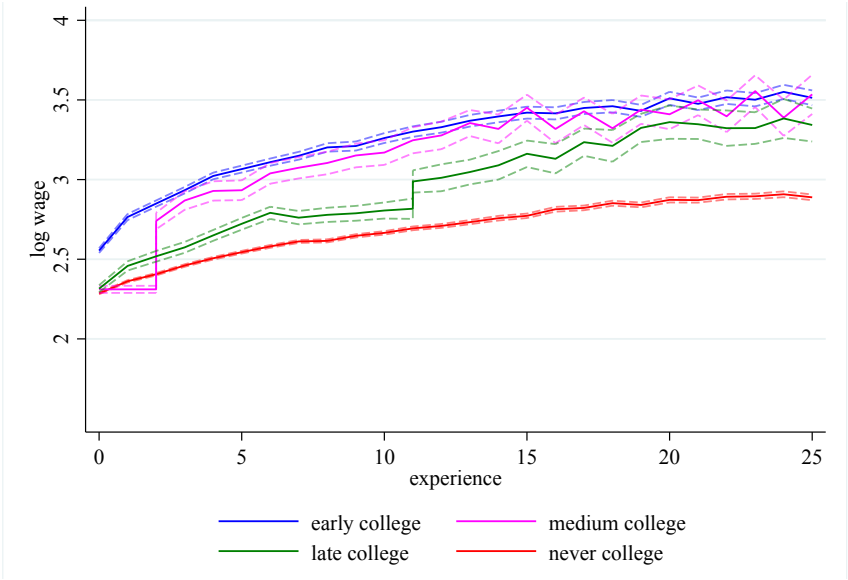
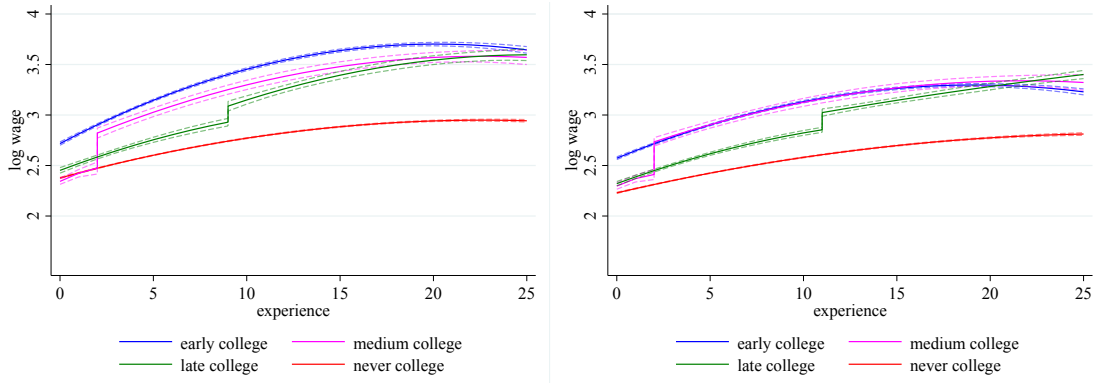


Figure C.11: Average wage trajectories by group

Notes: Authors' calculations from the NLSY79. Each curve shows the predicted wage path (and 95% confidence interval) against experience from regression (7) for the representative individuals in the four college groups.

C.5 College wage premium by gender and race

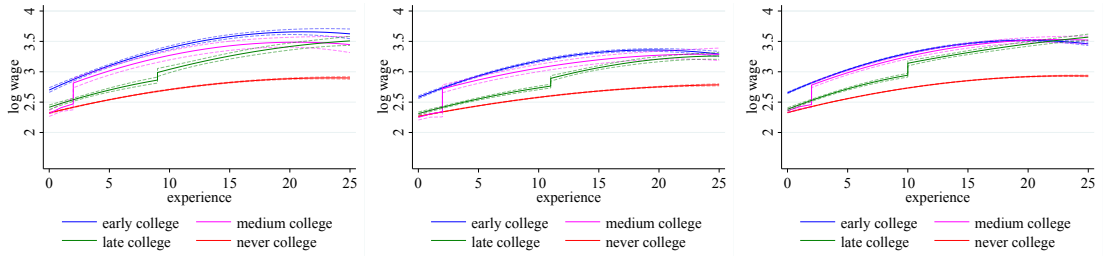
Figure C.12 shows the wage trajectory for the representative individual from each college group when equation (1) is estimated separately by gender on the NLSY79. The dynamics are somewhat different for women and men. Female early graduates' returns to experience are flatter and start lower than male early graduates'. Also, female medium graduates catch up with female early graduates, while this is not true of male medium graduates. Finally, male late graduates experience a steeper increase in their returns to experience after they obtain a college degree than female late graduates.
Figure C.13 shows representative wage paths for Hispanics, Blacks, and non-Hispanic, non-Black individuals in each of the four college groups from the NLSY79. We also observe disparities between races. The gap between medium and early graduates is widest among Hispanics, followed by Blacks. Also, Black late graduates accumulate



(a) Men (b) Women

Figure C.12: Average wage trajectories by college group for men and women

Notes: Authors' calculations from the NLSY79. Each curve shows the predicted wage path (and 95% confidence interval) against experience from regression (1) for the representative individuals in the four college groups for men and for women.



(a) Hispanics (b) Blacks (c) others

Figure C.13: Average wage trajectories by college group and race

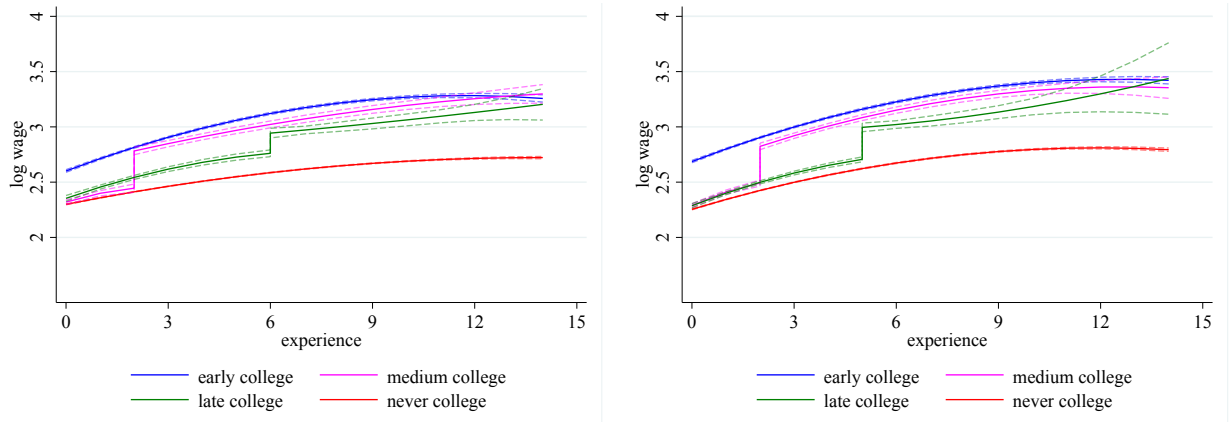
Notes: Authors' calculations from the NLSY79. Each curve shows the predicted wage path (and 95% confidence interval) against experience from regression (1) for the representative individuals in the four college groups by race/ethnicity.

more labor market experience than other races before obtaining a college degree, although it does not shift their wage paths higher once they obtain their college degree.

C.6 Wage paths in the NLSY97

To see whether the patterns look similar for the four college groups across different birth cohorts, we run similar wage regressions in the NLSY97 data. To be able to compare the regression results between the NLSY1997 and NLSY1979, we need to ensure that the wage trajectories are estimated over the same age span for both cohorts. For this reason we truncate the NLSY79 in 1998 in which year the age distribution in the NLSY79 survey (ages 31-41, with a mean of 36) is similar to the age distribution in the last year of the NLSY97 (ages 30-40, with a mean of 36). Figure C.14 shows the predicted wage trajectories based on the truncated NLSY79 in the left panel and based

on the NLSY97 in the right panel. Crucially, Figure C.14 indicates that the wage curves for the four groups are qualitatively very similar across the two samples.



(a) NLSY79 truncated (b) NLSY97

Figure C.14: Average wage trajectories in NLSY79 and NLSY97

Notes: Authors' calculations from NLSY79 and NLSY97 data. In each panel the curves show the predicted wage paths against experience from regression (1) as in Figure 5. The left panel shows these from the regression on the truncated NLSY79 data, while the right panel shows this from the NLSY97 data.

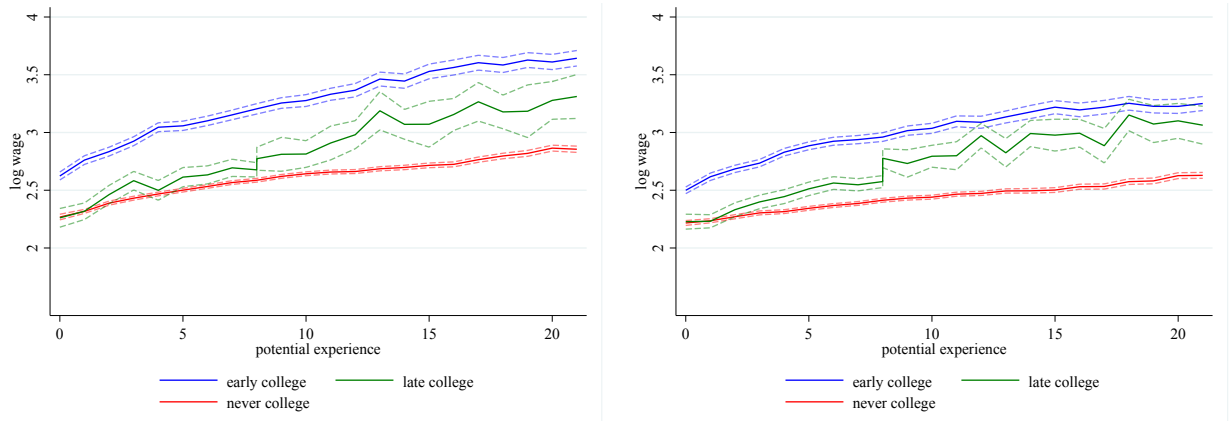
C.7 College wages for Section 4.2

In Section 4.2 we need measures of the average wage by age and gender of late bloomers and early graduates. We use this together with repeated cross-sectional data from the CPS ASEC, where we do not have any information on individual skills and on actual labor market experience. We obtain these wage measures from the NLSY79 by running the following regression by gender:

$$\log w_{it} = \alpha + \sum_{s=0}^{21} \beta_{s,n} \mathbf{I}\{\text{potexp}_{n,it} == s\} + \sum_{s=0}^{21} \beta_{s,c} \mathbf{I}\{\text{potexp}_{c,it} == s\} + \gamma' X_i + \epsilon_{it}, \quad (8)$$

for non-college workers, late bloomers (college obtained after age 29), and early graduates (college obtained up to age 29) separately. This equation is very similar to (7) except that instead of using actual experience, we calculate and use non-college and college potential experience, in order to calculate wages by age. Total potential experience is calculated as age minus years of schooling minus six, or zero if this is negative. College potential experience potexp_c is age minus the maximum between age at college graduation and 22, or total potential experience if that is smaller. Non-college potential experience potexp_n is total potential experience minus college potential experience, or zero if this is negative. The individual controls X_i only contain race dummies.

We compute the wage potential experience trajectory for each group by setting the race dummies' values at the group's averages, and by assuming that the college degree is obtained at the average of the maximum years of non-college potential experience across all individuals in the group. These trajectories are shown in Figure C.15 for men and for women separately. In Section 4.2, the values for $\log w_{g,a,ed}$ for early and late graduates are obtained from these graphs by mapping age to potential experience. We only need wages after graduation, and hence early and late graduates are assigned their respective college wage paths at potential experience $a - 22$.



(a) Men (b) Women

Figure C.15: Average wage - potential experience trajectories for men and women

Notes: Authors' calculations from NLSY79. In the two panels the curves show the predicted wage paths (and 95% confidence interval) against potential experience from regression (8), for men on the left and for women on the right.

D Data supplement for implications for the wage structure

D.1 Revisiting Card and Lemieux (2001)

We use the CPS until 2020 and follow Card and Lemieux (2001) as closely as possible in constructing relative college supply and college premium measures. Since earnings are reported from the previous year, we have data essentially until 2019. We create year groups 1980, 1985, 1990, 1995, 2000, 2005, 2010, 2015, where we use data from the 3-year window centered on the given year, and for 2019 we use data from 2018 and 2019. We also create 5-year age groups, 20-24, 25-29, etc. In each year group we calculate measures for individuals who were of the given age group age in the year 1980, 1985, etc., so the 25-29 age group in year 1985 includes all individuals who were 24-28 in 1984, 25-29 in 1985 and 26-30 in 1986. We follow Card and Lemieux (2001) and calculate the college premium based on a well-defined narrow sample, and calculate

the relative college supply based on a wide sample of individuals.
For the wage data, we keep only full time male workers, and workers with exactly a high school degree or exactly a college degree. We calculate hourly wages from yearly income divided by weeks worked times hours worked by week, and deflate wages. We regress log real hourly wages on a dummy for college, on a dummy for race, and on year fixed effects, and run the regression separately by year group and age group. The coefficient on the college dummy is our measure of the college wage premium, and we use the standard errors on these coefficient estimates as weights in the final regressions.
To construct high school equivalent and college equivalent labor supply by age group and year group, we aggregate annual hours supplied across different education groups weighted by the average wage (across all periods) of the given education group. High school equivalent labor of age group j in year t, H_{jt} is calculated as total annual hours worked by high school graduates in age group j, plus the total hours of high school dropouts (weighted by their wage relative to high school graduates), plus a share of the hours worked by workers with some college education. This share is the weight on the high school wage needed to express the wage of workers with some college education as a weighted average of high school and college wages. We calculate the amount of college equivalent labor of age group j in year t, C_{jt} as the total annual hours worked by college graduates in age group j, plus the total hours of those with more than a college degree (weighted by their wage relative to college graduates), plus the remaining share of the hours worked by workers with some college education.
We calculate the late bloomer share among those with exactly a college degree by age group and year group. We keep individuals born in 1935 or later, as we do not know the share of college graduates at age 29 for earlier born cohorts. Late bloomers by birth cohort and year is calculated as $\lambda_{g,c,t} - \lambda^{29}_{g,c,t}$, i.e., the difference between the share of those with exactly a college degree in the given year and when the cohort was age 29, and is zero up to age 29 of the given birth cohort. We aggregate these within age group and year group using sampling weights, and the late bloomer share is the ratio of late bloomers to those with exactly a college degree in the given year group and age group.

D.2 Gender pay gap and flattening college premium

We compute employment shares (person weights times hours worked times weeks worked) by gender, potential experience group among college workers for all years between 1985-2020. The fixed weights, $\overline{\omega}_{g,a,coll}$, are computed as the average of the yearly employment shares. These, together with $\eta_{g,a,t}$ and $\log w_{g,a,ed}$ are used to calculate the late bloomer adjustment of college wages.

The composition-adjusted college wage by gender in Figure 7 and the composition-adjusted college premium by gender in Figure 8 is computed using the CPS, following Autor et al. (2008).

In the CPS earnings are top-coded, we replace top-coded values with one and a half-times the threshold value. We compute weekly and hourly earnings from annual earnings and weeks and hours worked. We drop observations either with hourly wages below the federal minimum wage in that year or that exceed the top-coded value of annual earnings divided by 35 times 52. We transform nominal wages to real wages using the GDP deflator. We keep full time full year workers. We define 4 potential experience groups (0-9, 10-19, 20-29, 30+ years of potential experience) and 5 education groups (less than high school, high school, some college, college graduate, greater than college). Log real hourly wages are then regressed on the education category interacted with a continuous potential experience variable and its square, controlling for race (White, Black, other). The regression is run separately by gender and year, and is weighted by the labor share weights calculated as the person weights provided by the CPS multiplied by weeks worked. The mean log wage for each gender, education and potential experience category is the predicted log wage at the mid-point of the relevant potential experience group (5, 15, 25 or 35 years) for Whites.

The composition adjusted mean log non-college wage in each year is computed by taking the fixed-weighted average (similatrly to $\overline{\omega}_{g,a,coll}$, but using 5 education categoires) of the predicted log wages of less than high school, high school and some college individuals, over education and experience categories by gender of the given year. The composition adjusted mean log college wage is computed similarly from the college graduate and the greater than college categories. The college premium is the difference between the composition adjusted college and non-college wages. These are adjusted for the education and potential experience mix, but are not adjusted for the varying share of late bloomers.